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**STATISTICAL AND MACHINE LEARNING APPROACHES
TO EXPLORING BRAIN STRUCTURAL CHANGES IN
ALZHEIMER'S DISEASE PROGRESSION: A
LONGITUDINAL STUDY**

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**THIS PROJECT IS SUBMITTED TO THE DEPARTMENT OF STATISTICS
AND ACTUARIAL SCIENCE IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR THE AWARD OF THE BACHELOR'S DEGREE IN
STATISTICS**

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DECLARATION

We hereby declare that this project is the product of our own research and that none of it has been presented for credit toward any degree at this or any other university. All citations in this paper have been properly acknowledged.

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DEDICATION

We dedicate this research to the Almighty God, whose grace, strength, and guidance sustained us throughout this journey. To our families, friends, and loved ones, we extend our deepest gratitude for your unwavering support, constant encouragement, and immeasurable love. Your belief in us, your sacrifices, and your presence—whether near or far—gave us the motivation to persevere through every challenge. This work reflects your support, and we are forever grateful.

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ABSTRACT

Alzheimer's disease (AD) is one of the most significant contributors to cognitive decline and neurodegeneration in older adults. Timely intervention and management rely heavily on the early prediction of dementia progression. This research employed both statistical and machine learning methods to investigate the relationship between changes in brain structure and the progression of dementia, utilizing data from the Open Access Series of Imaging Studies (OASIS) longitudinal dataset.

Preliminary descriptive analyses were performed to examine the distribution of dementia severity across different groups. A one-way Repeated Measures MANOVA was used to evaluate the effects of group and time on cognitive and structural factors. Next, Principal Component Analysis (PCA) was applied to reduce dimensionality and identify key components that account for the most variance in the dataset. To address class imbalance related to dementia severity, the Synthetic Minority Oversampling Technique (SMOTE) was applied before training the predictive models. Five supervised machine learning methods—Gradient Boosting, Random Forest, Support Vector Machine, Neural Network, and Naïve Bayes—were developed and assessed using cross-validation and independent test datasets. The models' performance was evaluated with metrics such as accuracy, precision, recall, F1 score, and other relevant indicators.

Among all models, the Random Forest algorithm showed the highest predictive accuracy, effectively forecasting dementia progression and classifying subjects by severity. Age, MMSE, eTIV, and nWBV emerged as the key indicators of cognitive decline. The findings confirm the potential of machine learning to improve dementia diagnosis and progression modeling, while also supporting the use of neuroimaging and cognitive metrics for early detection.

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ABBREVIATION

AD.....	Alzheimer’s Disease
AdaBoost	Adaptive Boosting
ADNI.....	Alzheimer’s Disease Neuroimaging Initiative
AUC.....	Area Under Curve
BOLD.....	Blood Oxygenation Level Dependent
BIRN.....	Biomedical Informatics Research Network
CDR.....	Clinical Dementia Rating
CDR-SOB.....	Clinical Dementia Rating Sum of Boxes
CNN.....	Convolutional Neural Network
CRT.....	Cognitive Reserve
CSF.....	Cerebrospinal Fluid
CV.....	Cross-Validation
DMN.....	Default Mode Network
DMTs.....	Disease Modifying Therapies
F1	F1 score (Harmonic mean of precision and recall)
FAQ	Functional Activities Questionnaire
fMRI.....	Functional Magnetic Resonance Imaging
FN	False Negatives
FP.....	False Positives
FPR	False Positive Rate
GDS.....	Global Deterioration Scale
HHMI.....	Howard Hughes Medical Institute
IQ.....	Intelligence Quotient
LMICs.....	Low- and Middle-Income Countries

MANOVA.....	Multivariate Analysis of Variance
MCI.....	Mild Cognitive Impairment
MLP.....	Multi-layer Perceptron
MMSE.....	Mini-Mental State Examination
MRI	Magnetic Resonance Imaging
NC.....	Normal Control
NFT.....	Neurofibrillary Tangles
NRG.....	Neuroinformatics Research Group
OASIS.....	Open Access Series of Imaging Studies
PCA.....	Principal Component Analysis
RF.....	Random Forest
ROC.....	Receiver Operating Characteristic
SMOTE.....	Synthetic Minority Over-sampling Technique
sMRI.....	Structural Magnetic Resonance Imaging
SSA.....	Sub-Saharan Africa
SVM.....	Support Vector Machine
SVF.....	Semantic Verbal Fluency
TN.....	True Negatives
TP.....	True Positives
TPR.....	True Positive Rate
WHO.....	World Health Organization

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CHAPTER ONE

INTRODUCTION

1.0 Overview

This chapter discusses how Alzheimer's disease (AD) results in significant changes in the brain's structure and function. It examines the effect of neurodegeneration on the decline in the brain's proper functioning and highlights the importance of tracking these changes as they unfold. It further assesses the role of Magnetic Resonance Imaging (MRI), a brain imaging technique, in detecting neurodegeneration and studying the structural changes in the brain associated with Alzheimer's disease.

1.1 Background

Alzheimer's disease (AD), a neurodegenerative disorder that impairs cognition, memory, and behavior, has shown an alarming rise in the global public health sector. Alzheimer's disease is proven to be the primary cause of dementia globally. It is increasingly becoming one of the most devastating, troubling, and lethal diseases of the century, and the incidence of dementia is predicted to triple by mid-century (Scheltens et al., 2021). Despite considerable efforts to attain effective treatment for Alzheimer's disease therapy, drug development has not achieved therapeutic success. In 2023, the World Health Organization (WHO) reported that over 57 million people have dementia worldwide, and Alzheimer's disease accounts for approximately 60-70% of these cases, with over 60% living in low- and middle-income countries (LMICs). Interestingly, the most recent data shows that dementia increases by approximately 10 million new cases annually. This is chiefly due to the rise in life expectancy and global population growth.

In North America, the prevalence of dementia is connected chiefly with the aging population. Approximately 6-10% of persons 65 years and older have dementia, with Alzheimer's disease

(AD) contributing to about two-thirds of these cases. Alzheimer's Association (2024) reports that approximately 7 million Americans are living with Alzheimer's, and it is projected to rise to nearly 13 million by 2025, provided no crucial breakthrough to preventing and treating Alzheimer-related diseases is made. Not surprisingly, Alzheimer's disease was the fifth leading cause of death among Americans aged 65 years and above in 2021.

Uwishema et al. (2024) report that Alzheimer's disease poses an existential and escalating public health threat across Africa, affecting approximately 2.67 million people, with a greater percentage living in Sub-Saharan Africa. African studies on Alzheimer's conducted in 2012 reveal that the incidence of Alzheimer's disease in Africa is 2.4%, predicted to rise drastically by mid-century, primarily driven by the aging population, increasing life expectancy, and significant risk factors, including low education, hypertension, smoking, and obesity (George-Carey et al, 2012)

In 2015, approximately 2.13 million people in Sub-Saharan Africa (SSA) were living with dementia. The number is expected to rise to about 3.48 million in 2030 and 7.62 million in 2050, respectively. A larger percentage of the predicted increase is likely to come from the Central and Eastern regions of the SSA, driven primarily by the rising aging population. In 2015, over 367,000 new cases of dementia were recorded (Guerchet et al., 2017).

Over the past few years, the incorporation of neuroimaging techniques has reenergized efforts to examine the structural brain changes associated with Alzheimer's. Magnetic Resonance Imaging (MRI) provides neurologists and researchers with detailed and non-invasive images of the periodic structural changes in the brain associated with Alzheimer's disease that may have gone unnoticed by human observers, for instance, the accumulation of β -amyloid plaques and neurofibrillary tangles (NFT), leading to neuronal damage and brain atrophy. A longitudinal MRI study would enable the analytic understanding of dynamic changes in the

brain over time, including neurodevelopment and disease progression, making it essential for understanding cognitive and behavioral changes.

This research aims to analyze longitudinal MRI data to explore the brain's structural changes in individuals diagnosed with Alzheimer's. With the insights this study aims to reveal, researchers and neuroscientists will enhance their understanding of the disease's progression and develop resources for early detection and monitoring. This would align with ongoing initiatives to employ therapeutic strategies for improved healthcare outcomes in Alzheimer's disease.

1.2 Problem Statement

Alzheimer's disease continues to pose a significant challenge to the global healthcare systems, neuroscientists, and researchers. This is primarily due to the progressive nature of the disease and the current lack of a definitive cure, despite ongoing efforts to develop one (WHO, 2025).

Alternative approaches have been sought to analyze and predict the onset and prevalence of Alzheimer's disease. A study by Taeho Jo et al. (2019) employed a deep learning approach, along with a combination of deep learning and traditional machine learning, for classification. The findings of their study resulted in about 98.8% precision for Alzheimer's disease classification and an 83.7% efficiency for predicting mild cognitive impairment (MCI) conversion to Alzheimer's disease. Similarly, a combination of multimodal neuroimaging and fluid biomarkers is expected to yield better classification performance.

Garam Lee et al. (2019) employed a deep learning approach to monitor Alzheimer's disease progression by integrating a framework that combines cross-sectional brain imaging, longitudinal cerebrospinal fluid (CSF) biomarkers, and cognitive performance biomarkers. In addition, the outlined approach embedded multi-domain data to enhance the prediction

accuracy. A prediction precision of 75% was achieved using a single data modality. Nevertheless, when longitudinal multi-domain data were employed, the prediction accuracy improved to 81%, resulting in a six-percentage-point enhancement.

Liu et al. (2014) employed deep learning techniques to propose a method for the timely diagnosis of Alzheimer's disease, which concurrently reduces dimensionality and data fusion to preserve the unified effort between data modalities. When evaluated against a support vector machine (SVM) classifier, the system was observed to achieve a performance advantage in terms of overall accuracy and specificity

Despite advancements in deep learning techniques for predicting Alzheimer's, significant gaps still exist. The use of longitudinal structural MRI data to model disease progression is limited; reliance on multimodal datasets that are not widely accessible persists, and there is a lack of effective integration between dimensionality reduction methods and classification models for dementia severity, which increases the risk of overfitting. This research addresses these problems by employing PCA-based feature extraction along with various machine learning algorithms to predict and classify the progression of Alzheimer's disease based on a purely structural and demographic longitudinal dataset.

1.3 Research Objectives

The primary objective of this study is to utilize longitudinal MRI data to investigate and examine the progressive brain structural changes and cognitive decline associated with Alzheimer's disease.

The specific objectives subsume;

1.3.1. To classify and predict subjects on dementia severity.

1.3.2. To investigate whether there is a connection between dementia status and combined observable variations in brain structure and cognitive functions.

1.4 Significance of the Study

This study provides valuable insights to the global health community by delivering scientific and medical information essential for understanding the gradual effects of Alzheimer's disease on brain anatomy throughout its stages. Such knowledge facilitates early detection, leading to prompt intervention, enhanced patient care, and accelerated development of therapeutic strategies.

The proliferating aging of the population, associated with the rising number of people with Alzheimer's disease, and the urgent need to document prognosis, trigger the study of the progressive structural brain changes. Longitudinal research is crucial for elucidating the nature of AD. Ultimately, it enables researchers, neurologists, and clinicians to distinguish clearly between the stages and subtypes of Alzheimer's disease, as noted by Jerome et al. Applying machine learning algorithms to longitudinal MRI data enhances the timely identification of high-risk individuals for dementia by addressing potential limitations of human assessment, such as uncovering hidden and complex patterns in the data.

This research aims to benefit as many users as possible. Based on these findings, healthcare systems could enhance the accuracy of AD detection, leading to targeted therapy and improved patient outcomes. MCI subjects stand to gain from the findings of this research, which can help them better protect themselves from progressing into AD. Similarly, persons diagnosed with AD, as well as caregivers, could benefit from the earlier detection and mitigation strategies by the healthcare systems to decelerate the progression of the disease, thereby reducing the risk of dementia diagnosis. Neuroscientists and researchers could also benefit from the findings of

this study, as they provide valuable insights into the relevance of biomarkers (such as MRI scans) in AD progression research, potentially leading to further studies.

1.5 Brief methodology

This study would employ secondary data analysis. The data used in this research were procured from a database available at <https://www.kaggle.com/datasets/jboysen/mri-and-alzheimers/data>. The Open Access Imaging Studies (OASIS) initiative aims to provide the scientific community with a freely available MRI dataset to enhance future discoveries in basic and clinical neuroscience. Included in the OASIS dataset are: Cross-sectional MRI Data in Young, Middle Aged, Nondemented and Demented Older Adults, and Longitudinal MRI Data in Nondemented and Demented Older Adults. This study focused on longitudinal MRI Data in Non-demented and demented older adults. The set constituted an initial sample size of 150 subjects aged 60 to 96. Multiple scans, with an interval of at least one year, were performed on these subjects, totaling 373 imaging sessions. From the initial 150 subjects, 72 and 64 remained non-demented and demented, respectively, throughout the imaging sessions. Fourteen subjects were, in their initial scans, labeled as non-demented, but later scans characterized them as demented.

The statistical approaches and machine learning methods employed in this research include Multivariate Analysis of Variance (MANOVA), Principal Component Analysis (PCA), and other classification methods that may evolve as the study progresses. These data analysis techniques would enable the concurrent assessment of differences in MRI variables between demented and non-demented subjects, allowing the transformation of a large set of variables into smaller sets without omitting the substantive information in the original data.

1.6 Tools for Data Analysis

The primary tool used in this study will be the R statistical software. The selection of R for this study is based on its extensive range of powerful packages for statistical modeling, data visualization, and machine learning algorithm development, all of which are suitable for analyzing the OASIS MRI longitudinal data. This enables the acquisition of insights into data trends and patterns, allowing for the making of appropriate statistical inferences.

1.7 Chapter Organization

This study is structured into five chapters, each serving a targeted purpose in exploring the structural brain changes associated with Alzheimer's disease.

Chapter one introduces the topic by providing a brief overview of Alzheimer's disease, highlighting its significance as a global health issue and how the brain anatomy responds to the progression of the disease. It also presents the problem statement, research objectives, methodology, and the tool for analyzing the data.

Chapter two reviews existing relevant literature on Alzheimer's disease, examining previous research findings, the theoretical framework, and empirical inspection that inform the relationship between Alzheimer's disease and the structural brain changes. This review will outline the gaps this study aims to address.

Chapter three details the methodology used in this research. It will explain the data source and acquisition method. Additionally, it will explain the variables in the OASIS longitudinal MRI data and the techniques utilized for data analysis.

Chapter four analyzes and presents the MRI data acquired. Descriptive statistics will initially summarize the data and then make inferences. Regression analysis and hypothesis testing will be performed to estimate the relationship between variables in the data.

Chapter five summarizes the key findings of the research, drawing conclusions based on the data analysis. Practical recommendations will also be drawn for public health and policy-making. It will provide suggestions for future studies that aim to address the setbacks encountered during this study.

CHAPTER TWO

LITERATURE REVIEW

2.0 Overview

Alzheimer's disease is primarily referred to as a brain defect that affects memory and thinking skills, ultimately causing neurocognitive disorders. On average, AD commences after 60 and may persist for 8 to 12 years. There are two major types of Alzheimer's disease: early-onset Alzheimer's disease and late-onset Alzheimer's disease. Early-onset Alzheimer's disease commences in advance of age 65 and is considered to be chiefly hereditary, although not phenotypically expressed (Ridge et al., 2013). Late-onset Alzheimer's disease begins post age 65 and is the most common form of Alzheimer's, contributing to approximately 90-95% of the gross Alzheimer's cases globally. While both types progress, early-onset AD progresses rapidly and exhibits swift cognitive deterioration.

2.1 History of Alzheimer's Disease

Auguste was born to the Kassel family in Germany in the mid-19th century. At age 23, she married Karl Deter and later cohabited with him in Frankfurt. Twenty-eight years later, Auguste's life became increasingly chaotic, full of sudden accusations, hallucinations, and alarming neighbors with a terrifying scream. Her mental health declined swiftly, and she began to mistrust the actions of unfamiliar individuals without cause.

Owing to the rapidly deteriorating condition of Auguste's cognition, Karl, his husband, transported her to a mental hospital for medical attention. A medical practitioner named Dr. Alois Alzheimer closely monitored her condition. Following a thorough review of the patient's history, Alzheimer determined that the case was a form of "presenile" dementia and requested to own the patient's clinical history and brain after her demise (Yang et al., 2016). Soon after

Dr. Alois Alzheimer's exit from the mental hospital, Mrs. Auguste Deter's condition worsened significantly, causing her inability to perform daily tasks, including eating. On April 8, 1906, four years after Alzheimer's departure, Mrs. Auguste passed away, following her prolonged illness related to septicemia and pneumonia due to blood poisoning and pressure injuries, as revealed by the initial post-mortem examinations (Aminoff, 2014; Yang et al., 2016). The medical history, together with the brain of the deceased, was handed over to Dr. Alzheimer upon his earlier request.

Yang et al. (2016) report that Alzheimer conducted detailed research on the brain to track the connection between the clinical records and the symptoms that unfolded. He realized an atrophy in the cerebral cortex, responsible for disrupting the brain's higher-level functions, including thinking, language, and memory. The accumulation of plaques around the neurons and the development of tangles in the nerve fibers were also observed, which can be characterized as early-onset Alzheimer's disease in recent times. These findings were presented at a medical conference in April 1906 at Tübingen by Alois Alzheimer himself, although they initially failed to attract noticeable attention from distinguished scholars. In 1910, Dr. Emil Kraepelin officially coined the eponym "Alzheimer's disease" in the eighth revised edition of his *Clinical Psychiatry* book, acknowledging Dr. Alois Alzheimer's research findings in Auguste Deter's case (Bermudez, 2012; Yang et al., 2016).

Before the mid-20th century, Alzheimer's disease was considered to be infrequent, reserved for individuals before age 65 years, and termed as presenile dementia (White, 2011). Senile dementia was ascribed to dementia syndromes that develop after age 65 years and are thought of as part of normal aging. However, researchers in the late 1960s discovered that Alzheimer's disease ignores the age restriction placed on it, as the same neuropathologic lesions that distinguish presenile dementia were seen to be present in individuals above the age of 65 years.

This resulted in the reclassification of Alzheimer's disease as the most typical cause of dementia in individuals, regardless of age group (White, 2011; Middleton, 1986).

The late 20th and early 21st centuries marked a period of rapid development in inspecting the neurobiological framework, as well as the diagnostic and therapeutic focus of the disease. Beta-amyloid, $A\beta$, was discovered as the main component of amyloid plaques in Alzheimer's disease in 1984 (Tien-Phat & Holtzman, 2018). Kosik et al. (1986) report that tau-protein, τ , is a primary antigenic component of neurofibrillary tangles, **NFTs**, which are associated with the disease. The unveiling of neuroimaging techniques through the establishment of the Alzheimer's Disease Neuroimaging Initiative (ADNI) in 2004 aimed to identify the relationships among the clinical, cognitive, and genetic features associated with the disease's spectrum (Weiner et al., 2010). These discoveries were instrumental and remain the most promising standardized medical protocols for the timely detection and disease progression, as there is no current definitive cure for Alzheimer's disease.

Initial statistical evaluations of the disease focused on descriptive statistics, correlational studies, and basic regression methods to measure its prevalence and connect cognitive assessment results, clinical demographics, and brain-derived characteristics to neuropathological findings. The earliest comprehensive statistical analysis on Alzheimer's disease seems to be a study from 1993 conducted by Azari et al., which utilized positron emission tomography (PET) data alongside sophisticated statistical methods to explore discriminant functions that classify AD patients from controls, achieving an 87% classification accuracy (Azari et al., 1993). Their cross-sectional study was limited by a smaller sample size, which challenged the generalizability of the results, and a suggestive result.

Foundational works on longitudinal cohort studies, alongside the implementation of neuroimaging data, closely followed those on cross-sectional studies in the late 20th century

through to the early 21st century, focusing on growth and survival analysis in modeling rates of cognitive decline and disease progression. The work of McArdle et. al (2005) was central to this. In their study, new statistical methodologies were employed, utilizing longitudinal data to estimate survival and quantify changes in the early prediction of the disease, yielding optimal predictors of impending dementia (McArdle et. al, 2005). The limited sample size again affects the precision of the estimates for the shared-parameter models.

Over the past twenty years, studies on Alzheimer's disease have increasingly focused on statistical learning and machine learning methods, such as Random Forest, SVM, Gradient Boosting, Neural Networks, and ensemble approaches, which deliver improved predictive accuracy for both identifying the disease and forecasting its progression.

2.2 Theoretical framework

2.2.1 Stages of Alzheimer's Disease

Alzheimer's disease evolves progressively, with the pathophysiological changes commencing many years before the clinical symptomology develops along a spectrum. This earliest phase of the disease, before the clinical manifestation, is termed preclinical Alzheimer's disease. According to the Alzheimer's Association, an Alzheimer's patient's life spans, on average, four to eight years after diagnosis, although one can live as long as 20 years. Despite the patient-specific, progressive, non-linear, and multifaceted nature of AD, various staging models have been established to classify the disease into discrete, distinct, and defined clinical stages, which are critical for developing robust disease-modifying therapies (DMTs).

Early Stage (Mild) Alzheimer's

The early stage is the first clinical phase of AD and precedes the clinical diagnosis. This stage is characterized by pronounced memory deficit. At this stage, the patient experiences reduced

social functions, may encounter difficulty with language, and undergoes changes in behavior and mood, which typically last approximately three to four years. However, the individual performs uninterrupted daily tasks independently. Whitwell et al. (2007) report that atrophy in the hippocampus becomes visible in this stage through brain imaging findings, resulting in a volume depletion of the brain region responsible for memory and learning. During this stage, symptoms may only be detectable to the patient and their closest individuals (Butler Hospital, 2023; Alzheimer's Association, 2025). According to Butler Hospital (2023), obtaining a diagnosis in this stage enhances improved planning and management throughout the progression of the disease.

Middle Stage (Moderate) Alzheimer's

The middle stage is generally the longest in duration of AD and may persist for years. During this stage, the Alzheimer's patient grows more confused and forgetful. Patients in this stage are unable to perform social functions independently and may require assistance with basic daily tasks. The Alzheimer's Association reports that, with the growing demand for more intensive care for Alzheimer's patients in this stage, caregivers might explore options like temporary care services to provide a safe and supportive environment for the patient. This allows caregivers a brief respite while ensuring the patient receives the necessary care.

According to a report by Mayo Foundation for Medical Education and Research (2023), the affected individual may wander, perhaps searching for a more recognizable environment, as disorientation in time, place, and unfamiliar environment sets in. The brain structure undergoes significant deterioration at this stage as the disease progresses throughout the brain. Atrophy of the brain continues progressively, outstretching to the parietal lobe, as well as thinning of the cerebral cortex (Alzheimer's Society, 2021).

Late Stage (Severe) Alzheimer's

A significant cognitive deficit characterizes the final stage, and the patient may require intensive around-the-clock care (Clifford et al., 2023; Alzheimer's Association, n.d.). The AD-diagnosed individual in this stage loses physical ability, hence becomes entirely dependent on caregivers to be able to carry out basic daily activities such as walking and maintaining personal hygiene. The patient becomes vulnerable to infections, predominantly aspiration pneumonia, due to difficulty in swallowing and reduced ambulation (Alzheimer's Association, n.d.). Complete disorientation of the environment, as well as a decline in the patient's conventional communication, is welcomed in this stage. The brain undergoes volumetric atrophy and extensive damage, disrupting approximately its entire function.

2.2.2 Cognitive Theories and Models of Decline

Theoretical models of cognitive decline focused on understanding and appreciating how Alzheimer's disease alters thought patterns and behavior have been of crucial concern. These models seek to underscore the interplay between the biological, psychological, and social factors of cognitive decline, aiding in explaining the relationship between brain pathology and AD. In essence, these frameworks enhance our appreciation of neuroimaging data and forecast the disease's trajectory. These theories include Cognitive Reserve Theory (CRT), Disconnection Theory, and models connecting cognition with brain structural changes.

- **Cognitive Reserve Theory**

The Cognitive Reserve Theory posits that the brain makes an effort to manage brain damage before clinical manifestation (Stern, 2007). This model attempts to explain the individual's varying capacities in surviving brain pathology. The individual variability in cognitive reserve can be attributed to genetic differences or life experiences, such as

educational attainment, intellectually demanding occupational experiences, leisure activities, and other social engagements (Stern, 2007; Tucker et al., 2014). Stern et al. (1994) and Tucker et al. (2014) report that these variables tend to be positively related to cognitive reserve and better clinical outcomes; thus, higher levels of education, occupational attainment, and intelligence quotient (IQ) are accurate forecasters of high cognitive reserves. The theory presupposes that the reserve utilizes alternate brain networks and strategies to maintain performance amidst brain damage. Hence, there is no explicit relationship between the degree of brain pathology and observed cognitive functional deficits.

- **Disconnection Theory**

The disconnection theory proposes that cognitive decline associated with Alzheimer's disease results from the inability of the brain's communication networks to interconnect optimally, rather than solely from focal neuronal loss (Bokde et al., 2009). Sanz-Arigita et al. (2010) report that the unusual reorganization of unimpaired functional connectivity networks results in cognitive decline in AD patients. The progressive demyelination of the white matter and the damage to the synaptic connections disrupt the effective transmission of information across the brain regions, which plays a crucial role in the pathogenesis and diagnosis of the disease (Sachdev et al., 2013). According to Jones et al. (2011), reduced connectivity in brain networks, such as the default mode network (DMN), accounts for the low activity observed during internally focused mental states.

- **Models Linking Cognition with Brain Structure**

Various models have been proposed to explain how brain structure supports cognitive and functional processes. These models link cognitive decline to the brain structural changes.

The **Sequential Model of Decline** posits that Alzheimer's disease follows a consistent pattern of cognitive deterioration; thus, several staging systems describe the disease's predictable progression. Hennekes et al. (2016) identify the sequence of cognitive decline in AD patients as starting with impairment in recall, followed by impairment in orientation in time, attention, and concentration, and then in orientation in place, language, drawing, and registration.

The **Atrophy-Cognition Model** suggests that the progressive neurodegeneration in distinct brain regions of an AD patient corresponds to a gradual decline in cognition. Thus, as the disease progresses, measurable shrinkage of notable brain regions, such as hippocampal atrophy, results in equivalent cognitive symptoms.

The **Biomarker-Cognition Model** incorporates recent diverse techniques in understanding how diagnostic indicators of Alzheimer's disease associate with observable cognitive symptoms, subsuming episodic memory loss, among others. Jack et al. (2010) propose a model associating the stages of the disease with AD biomarkers: β -amyloid ($A\beta$) biomarkers become abnormal first, followed by neurodegenerative biomarkers and neuropsychological manifestations that connect with clinical symptom severity.

2.2.3 MRI as a Tool for Studying Neurodegeneration

Magnetic Resonance Imaging (MRI) has been identified as a high-powered tool for studying neurodegeneration, offering detailed structural and functional non-invasive images of the brain in investigating the gradual alterations of the brain as the disease progresses. Huang & Zhang (2022) report that MRI has been extensively utilized in both animal research and large-scale clinical investigations, providing a linkage between micro- and macro-level analysis and promoting bench-to-bedside translational research due to its non-invasive and non-radioactive nature. In Alzheimer's research, the two basic types of MRI used are structural MRI (sMRI) and functional MRI (fMRI).

Structural Magnetic Resonance Imaging (sMRI)

Structural magnetic resonance imaging is a neuroimaging tool used to visualize and quantify the brain morphometry; it produces high-resolution, detailed 3D images of the brain and detects gray matter atrophy, cortical thinning, and ventricular enlargement linked to neuronal and synaptic losses that occur on a microscopic level in AD (Vemuri & Jack Jr, 2010). Volumetric variations of the distinct brain regions associated with AD progression are also measured by sMRI scans, facilitating comparison among AD patients as well as the disease's stages.

Functional Magnetic Resonance Imaging (fMRI)

Functional magnetic resonance imaging, on the other hand, is a non-invasive tool used to visualize brain activity and network connections. Buxton (2013) reports that fMRI is used to detect the dynamic patterns of activity in the active human brain. The fMRI method mainly relies on blood oxygenation level-dependent (BOLD) contrast, which is based on the contrasting magnetic properties of oxygen-rich and oxygen-poor blood (Buxton, 2013; Noll, 2001). Shah et al. (2010) note that fMRI plays an increasingly important role in preoperative

planning by identifying eloquent brain regions in relation to intracranial pathologies, as well as enhancing our understanding of the neural basis of significant psychiatric disorders, neurological conditions, and substance use disorders.

2.3 Empirical Framework

2.3.1 Brain Atrophy Patterns and Progression in Alzheimer's disease

A substantial amount of research has utilized neuroimaging techniques to explore the structural brain alterations associated with Alzheimer's disease (AD). These investigations have been conducted using both cross-sectional and longitudinal designs, which enable researchers to assess the current state of the disease (cross-sectional) and track its impact on biomarkers and brain anatomy over time as the disease progresses (longitudinal).

Cross-sectional studies of AD have revealed appreciable progressive brain deformation, including hippocampal shrinkage, cortical thinning, and ventricular enlargement, when compared to age-matched healthy controls. For instance, Apostolova et al. (2012) conducted a cross-sectional study of Alzheimer's disease on MRI data from 46 cognitively normal elderly individuals (NC), 33 participants with mild cognitive impairment (MCI), and 43 AD subjects to analyze the hippocampal and ventricular volumes using two innovative semi-automated segmentation methods accompanied by the radial distance mapping techniques. In their study, multiple regression was employed to evaluate the impact of age and diagnosis on hippocampal and ventricular volumes, as well as radial distance. Their findings suggested that the most pronounced hippocampal atrophy and ventricular enlargement were observed in the comparison between AD and NC. Subjects with MCI exhibited intermediate levels of both hippocampal atrophy and ventricular enlargement. Significant age-related effects on hippocampal volume and radial distance were identified in the combined sample, as well as in the NC and AD groups separately (Apostolova et al., 2012). Jack Jr. et al. (2000) analyzed

cross-sectional data on 129 subjects from the Mayo Clinic AD Research Center who underwent MRI examinations of the head, recording the annualized percentage change in hippocampal volume. Findings from Jack Jr. et al. (2000) highlight that the annualized rates of hippocampal volume reduction for the three initial clinical groups showed a progressive decline in the following order: AD > MCI > NC. Among the control and MCI groups, those who experienced a decline had a significantly higher rate of volume loss compared to those who remained clinically stable. The average annualized rates of hippocampal atrophy by follow-up clinical group were as follows: control-stable (1.73 percent), control-decliner (2.81 percent), MCI-stable (2.55 percent), MCI-decliner (3.69 percent), and AD (3.5 percent). The strength of cross-sectional research lies in its efficiency and reduced expense; however, its drawback is the inability to track changes over time, which can lead to ambiguity regarding whether the observed differences indicate actual disease progression or existing anatomical variability.

Longitudinal studies of MRI research have offered an important understanding of brain atrophy patterns throughout the spectrum of Alzheimer's disease, as it tracks changes in the progressive deformation of the brain. Morra et al. (2009) undertook a longitudinal analysis of the 3D pattern of hippocampal degeneration over a period of time in 490 participants who underwent two brain MRI scans spaced a year apart (totaling 980 scans). In their work, Morra et al. (2009) evaluated the initial and one-year follow-up scans for 97 individuals with Alzheimer's disease (49 males and 48 females), 148 healthy control participants (75 males and 73 females), and 245 individuals with mild cognitive impairment (MCI; 160 males and 85 females). The Adaptive Boosting algorithm (AdaBoost) was employed to generate 3D models of the hippocampal surface from all 980 scans. Results revealed an increased rate of hippocampal volumetric loss with the severity of diagnosis (normal = 0.66%/year; MCI = 3.12%/year; AD = 5.59%/year) and was associated with both initial and periodic changes in Mini-Mental State Examination (MMSE) scores, as well as global and sum-of-boxes Clinical

Dementia Rating scale (CDR) scores. Additionally, surface-based statistical maps illustrated a distinct pattern of ongoing atrophy across all three diagnostic categories (Morra et al., 2009). Similarly, using a longitudinal approach, Sluimer et al. (2008) investigated the brain of 65 patients with Alzheimer's disease (38 females, 27 males; aged 52 to 81 years), 45 patients with mild cognitive impairment (22 females, 23 males; aged 56 to 80 years), 27 individuals with subjective complaints (12 females, 15 males; aged 50 to 87 years), and 10 healthy controls (six females, four males; aged 53 to 80 years), with two MRI scans obtained at a mean interval of $1.8 \text{ years} \pm 0.7$ (standard deviation) used in assessing the baseline brain volume and rates of whole-brain atrophy. Findings from Sluimer et al. (2008) indicate that brain volume was minimal in AD patients, though not significantly different from MCI groups, subjective complaints, and healthy controls ($p > 0.38$). As expected, Sluimer et al. (2008) realized a greater whole-brain shrinkage in AD patients ($-1.9\% \text{ per year} \pm 0.9$), followed by the MCI patients ($-1.2\% \text{ per year} \pm 0.9$), with MCI patients exhibiting higher atrophy rates than those with subjective complaints ($-0.7\% \text{ per year} \pm 0.7$) and controls ($-0.5\% \text{ per year} \pm 0.5$). These provide valuable insights into disease progression and patterns of change, enabling researchers to identify environmental and genetic factors that increase AD risk and contribute to cognitive decline. Global surveys have revealed that these studies often face considerable methodological challenges (unadjusted heterogeneity in demographic factors such as education and age between subjects, attrition, selective survival, and neuropsychological assessment error) that compromise research validity.

2.3.2 Cognitive Decline Assessment

Considerable cognitive evaluation tools, such as the Mini-Mental State Examination (MMSE) and the Clinical Dementia Rating (CDR), are used to screen and classify the disease into its distinctive, non-overlapping stages and track the disease's progression. Earlier publications

have demonstrated a substantial correlation between brain structural deformation and MMSE and CDR scores. Mouton et al. (1998) assessed a subset of 46 subjects on their performance on the Mini-Mental State Examination (MMSE). The report revealed that probable AD cases exhibited clinical signs of unexplained, severe, and progressive dementia (final MMSE score < 25 ; mean 6.7, SD 6.6), whereas the controls displayed no signs of neurological disease (final MMSE scores ≥ 25 ; mean 28.4, SD 1.5), indicating the presence of a strong correlation between cognitive function and regional brain volume (Mouton et al., 1998). Comparable results by Jack et al. (2002) discovered that AD patients with reduced hippocampal volumes had lower scores on the MMSE ($r = 0.60$), suggesting a direct connection between atrophy in the medial temporal lobe and memory deficits. Grundman et al. (2002) conducted a correlational analysis between the MRI hippocampal volumes recorded at the start of the study and the subjects' current performance on neuropsychological assessments and clinical evaluations. The analysis suggested that larger hippocampal volume correlated with improved performance on memory tests, overall cognitive measures, and general clinical assessments. It also indicated that a lower hippocampal volume at baseline might be related to an increased risk of progressing to clinical AD. Grundman et al. (2002) concluded that MRI hippocampal volume could serve as a valuable indicator of disease severity in MCI individuals and may act as a prognostic measure for the future development of AD.

Statistical and machine learning models have been established to forecast and analyze cognitive assessment tools, including MMSE, CDR, and CDR Sum of Boxes (CDR-SOB) scores, based on initial MRI characteristics. Linz et al. (2017) employed regression models on over 179 AD patients with degrees of dementia to predict MMSE and CDR scores using baseline Semantic Verbal Fluency (SVF) performance. The analysis from Linz et al. (2017) recorded a mean absolute error of 2.2 for the MMSE (with a range of 0–30) and 1.7 for the CDR-SOB (with a range of 0–18). The correlation between actual and predicted scores showed a Cohen's κ of

0.76 for MMSE and 0.52 for CDR-SOB (Linz et al., 2017). Sato et al. (2024) utilized baseline data from 944 participants in the Alzheimer's Disease Neuroimaging Initiative (ADNI) and 171 J-ADNI amyloid-positive participants, focusing on Mini-Mental State Examination (MMSE) and Functional Activities Questionnaire (FAQ) scores to establish a machine learning model for forecasting variations in CDR-SOB. The model by Sato et al. (2024) attained the necessary prediction accuracy ($MAE < 1.0$) and demonstrated satisfactory predictive performance ($R^2 > 0.5$) for patients with mild cognitive impairment (MCI) regarding changes in CDR-SOB over periods of 18 months or longer. Moreover, Perneczky et al. (2006) formulated MMSE scores in relation to CDR stages. Cohen's measure of agreement demonstrated a significant agreement for mild ($\kappa = 0.62$), moderate ($\kappa = 0.69$), and severe dementia ($\kappa = 0.76$), justifying their proposition that MMSE could be used as a substitute measure for CDR staging.

Nonetheless, several gaps are apparent in the existing literature. The lack of integration of MRI data with longitudinal cognitive trajectories, inadequate comprehensive model validation, and the crucial question of how correlations are assessed raise concerns about the robustness and accuracy of these predictive models, as well as their clinical relevance.

2.3.3 Machine Learning Applications in Alzheimer's Research

The swift emergence of machine learning algorithms as a ground-breaking resource for examining neuroimaging data to address the complex diagnostic and forecasting challenges in Alzheimer's research has raised some alarms. The far-reaching application of machine learning techniques has been employed to classify AD patients from healthy controls automatically and to establish predictive models for forecasting the conversion of MCI to AD. Multiple machine learning algorithms are featured prominently in the existing literature. The Support Vector Machine (SVM) algorithm has been applied extensively on an MRI dataset for AD

classification and regression tasks. Kruthika et al. (2017) utilized a support vector machine (SVM), accompanied by an sMRI data-derived anatomic brain features from 360 subjects processed by a FreeSurfer, to classify AD stages consisting of 130 standard controls, 150 mild cognitive impaired patients, and 80 AD patients, all sourced from Alzheimer's Disease Neuroimaging Initiative (ADNI). Publication by Yepes-Calderon et al. (2014) presents a straightforward approach for rapid diagnostic classification in clinical settings utilizing Support Vector Machines (SVM) and easily obtainable geometrical measurements, combined with cortical and sub-cortical brain parcellation. The classification model achieved classification success rates of up to 99.2% from a single measurement using an extensive imaging dataset comprising more than 800 subjects from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. Furthermore, the Random Forest (RF) is a widely used supervised machine learning algorithm that has been extensively applied in neuroimaging to manage complex, high-dimensional brain data in Alzheimer's classification. Song et al. (2021) tested the classification accuracies of random forest and other machine learning algorithms (Support Vector Machine (SVM), Multi-layer Perceptron (MLP), and Convolutional Neural Network (CNN)) with regional volumes of 2250 subjects from the ADNI database. The RF model achieved a classification accuracy of 90.2%, ensuring stability even when feature sets are limited. This suggests that RF forecasts AD more consistently with a reduced number of features (Song et al., 2021). Of particular relevance, the RF algorithm provides an accurate auto-calibrated framework for selecting a small set of jointly informative regions due to its high predictive accuracy and the use of variable importance measures (Genuer et al., 2010; Fabris et al., 2018).

It is essential to acknowledge specific limitations that hinder the advantages of using machine learning algorithms compared to traditional statistical methods. The neglect of significant clinical and demographic factors is prevalent throughout the literature. This lack of integration

restricts the model's generalizability and its application in real-world scenarios. Furthermore, employing a limited dataset leads to overfitting the model, which compromises its reliability.

2.4 Summary

This chapter has reviewed the historical and theoretical foundations, as well as the empirical studies connected to Alzheimer's disease. The literature heavily demonstrated how the disease progressively deteriorates the brain structure using the notable magnetic resonance imaging technique. Cross-sectional studies of Alzheimer's Disease (AD) have offered significant insights into understanding the disease's prevalence, risk factors, and its effects on brain structure. However, these studies often do not track disease progression, which limits their ability to establish causal relationships because they are based on one-time assessments. Longitudinal studies provide a more comprehensive understanding by allowing for the observation of the disease over time, thus facilitating the establishment of causal links.

The empirical framework highlights the connection between structural brain changes related to AD and cognitive assessments, such as the Mini-Mental State Examination and Clinical Dementia Rating scores. The substantial correlation documented in the literature supports the use of these clinical assessment tools as alternative biomarkers for gauging disease severity. The introduction of various machine learning algorithms for automating AD classification has further enhanced diagnostic and prognostic capabilities related to the progression of AD.

The existing literature reveals significant methodological weaknesses in AD research, particularly the failure to incorporate clinical demographic variables in predicting AD classification models. Additionally, the use of relatively small sample sizes in previous studies restricts the applicability of their findings and limits clinical relevance. By addressing these issues, this study seeks to build upon the current research by utilizing a large, well-defined

longitudinal dataset (OASIS) and applying robust machine learning algorithms to analyze changes in brain structure over time.

CHAPTER THREE

METHODOLOGY

3.1 Overview

The impact of Alzheimer’s disease (AD) cannot be given disproportionate weight. The social, economic, and psychological effects of the disease are not only borne by individuals but also by their families, caregivers, society, and the broader global community. Despite the diagnostic and prognostic advancements in medical practice for AD prevention and treatment, the worldwide control of AD is still far from optimum, as drug development to cure AD has not achieved therapeutic success. Thus far, various strategies have been developed to reduce behavioral risk factors, such as maintaining a nutritious diet, enhancing social and cognitive skills, promoting engagement and awareness, ensuring adequate sleep, avoiding smoking, and practicing self-management. This chapter describes the statistical modeling strategy and machine learning algorithms used to analyze longitudinal brain structural changes in Alzheimer’s disease, including the validation methods employed to confirm the reliability of the results.

3.2 Data Acquisition

The longitudinal magnetic resonance imaging (MRI) dataset from non-demented and demented older adults, made freely available to the scientific community by the Open Access Series of Imaging Studies (OASIS), is used for this study. This dataset comprises a longitudinal study of 150 individuals aged between 60 and 96. Each subject underwent at least two scans, with the scans spaced at least one year apart, resulting in a total of 373 imaging sessions. The OASIS collaborated with the Alzheimer’s Disease Research Center at Washington University, Dr. Randy Buckner from the Howard Hughes Medical Institute (HHMI) at Harvard University, the

Neuroinformatics Research Group (NRG) at the Washington University School of Medicine, and the Biomedical Informatics Research Network (BIRN) in making the dataset freely accessible to the neuroscience field. These organizations and prominent figures have dedicated research groups and resources for further investigations.

The dataset comprises multiple categorical and numerical variables, grouped into the following categories:

- **MRI-derived features:** Estimated Total Intracranial Volume (eTIV), Normalized Whole Brain Volume (nWBV), and Atlas Scaling Factor (ASF).
- **Clinical Demographic Variables:** Subject ID, MRI ID, Group (Demented or Nondemented), Visit, MRI Delay, Gender (Male or Female), Hand (Left or Right), Age, Education (EDUC), and Socioeconomic Status (SES).
- **Cognitive Evaluation Tools:** Mini-Mental State Examination (MMSE) score, and Clinical Dementia Rating (CDR)

Table 3.1: Dataset Composition

Variable	Description	Values / Categories
Subject ID	A distinct identifier for each subject in the dataset	Nominal (from OAS2_0001 to OAS2_0186)
MRI ID	The specific MRI session or scan identifier for each subject in the dataset	Nominal (from OAS2_0001_MR1 to OAS2_0186_MR3)
Group	This indicates whether the subject is part of a control group, has Alzheimer’s Disease (AD), or belongs to another specific group.	Demented, Nondemented, or Converted

Visit	This indicates the number of times a subject was scanned to track the changes over time.	Discrete (1-5)
MR Delay	This represents the time (delay) between different MRI scans or sessions on a subject.	Continuous (0-2639)
M/F	This denotes the gender of each subject.	Male or female
Hand	A specification on the dominant hand of the subject, though in this case, all subjects are described as right-handed	R
Age	This indicates the age of the subject at the time of the MRI scan	Continuous (60-98)
EDUC	Represents the years of education or the level of education completed by the subject	Discrete (6-23)
SES	A measure of a subject's economic and social position relative to others.	Ordinal (1-5)
MMSE	Mini-Mental State Examination score, a test to assess cognitive function	Continuous (4-30)
CDR	Clinical Dementia Rating, a scale used to quantify the severity of symptoms of dementia.	Continuous (0-2)
eTIV	Estimated Total Intracranial Volume, referring to the size of the intracranial cavity based on MRI data	Continuous (1106-2004)
nWBV	Normalized Whole Brain Volume, an MRI-derived measure that adjusts for total brain volume.	Continuous (0.644-0.837)
ASF	Atlas Scaling Factor, a metric used to normalize or standardize brain volumes to a standard atlas or template	Continuous (0.876-1.587)

This data aids in comprehending the causes, catastrophic effects, and toll of Alzheimer's disease. These variables are examined to identify and monitor health trends associated with Alzheimer's disease and to determine the factors that prompt further research. The data

collection involved participants who were right-handed and nearly evenly distributed by gender, with each individual having 3-5 distinct T1-weighted MRI scans collected during single sessions. The pre-processed MRI scans obtained were collated and cleaned with Microsoft Office Excel for further analysis.

3.3 Exploratory Data Analysis (EDA)

To thoroughly understand the longitudinal OASIS dataset and identify hidden relationships, trends, and statistical anomalies that could influence model implementation, an Exploratory Data Analysis (EDA) was performed. Distributions and potential outliers were illustrated using descriptive statistics and visual methods, including scatter plots, box plots, and histograms.

Descriptive statistics (minimum and maximum values, mean, mode, quantiles, median, and variance) were calculated to summarize the dataset and help identify potential outliers, ensuring that these values fall within reasonable ranges. For example, MMSE scores in the longitudinal OASIS dataset were cross-verified with standard MMSE scores. To enhance the visual representation of the qualitative factors in the dataset, the EDA employed univariate analysis on the variables. By organizing the data using tables, graphs, and summary statistics, the information was communicated clearly, and potential skewnesses were identified.

Bivariate analysis was performed to evaluate the connections between the predictors and the cognitive scores. For example, a bar chart was created to display the distribution of MMSE scores across different age groups. This allows for a quick examination of the degree and pattern of their relationship, thus facilitating conclusions about the age distribution of the population alongside their cognitive scores.

3.4 Statistical Modeling and Strategy

This study will primarily focus on the quantitative method of multivariate data analysis, including statistical techniques and machine learning models, to develop predictive models that utilize brain structural changes to forecast Alzheimer's disease progression and assess the impact of demographic features like age, education, gender, and socio-economic status on brain structure and cognitive decline. Furthermore, the discussion of results incorporates a qualitative interpretive aspect, describing the clinical significance of the identified brain regions and model outcomes within the context of existing research and the practical implications for early diagnosis and patient treatment.

3.4.1 Traditional Models.

Multivariate Analysis of Variance (MANOVA)

Multivariate Analysis of Variance (MANOVA) is a statistical technique that simultaneously compares the differences in group means across multiple dependent variables. By analyzing the variables together, MANOVA enables an interconnected analysis of correlated dependent variables. The null (H_0), and alternative (H_1) hypotheses are given as;

Null hypothesis (H_0): no significant difference in group mean vectors across the multiple dependent variables.

Alternative hypothesis (H_1): at least one of the group's mean vectors is significantly different.

The outcome of MANOVA is presented in any of the commonly used test statistics, called the Pillai's Trace $\mathbf{V}$, which expresses the trace of the Treatment (Between) Sums of Squares and the Cross-Products matrix $\mathbf{B}$, multiplied by the inverse of the sum of the Treatment (Between)

Sums of Squares and the Cross-Products matrix and the Error (Within) Sums of Squares and the Cross-Products matrix, $\mathbf{W}$.

$$\mathbf{V} = tr(\mathbf{B}(\mathbf{B} + \mathbf{W})^{-1}) \quad (3.1)$$

$$\mathbf{B} = \sum_{\ell=1}^g n_{\ell}(\mathbf{x}_{\ell} - \bar{\mathbf{x}})(\mathbf{x}_{\ell} - \bar{\mathbf{x}})' \quad (3.2)$$

$$\mathbf{W} = \sum_{\ell=1}^g \sum_{i=1}^{n_{\ell}} (\mathbf{x}_{\ell i} - \bar{\mathbf{x}}_{\ell})(\mathbf{x}_{\ell i} - \bar{\mathbf{x}}_{\ell})' \quad (3.3)$$

with $\mathbf{W}$, the Error (Within) Sums of Squares and the Cross-Products matrix, $\mathbf{B}$, the Treatment (Between) Sums of Squares and the Cross-Products matrix.

The Pillai's Trace $\mathbf{V}$ assesses the extent of variability that can be accounted for by the independent variables. It ultimately provides a value that helps determine whether to support or reject the null hypothesis, ranging from 0 to 1. A small value of $\mathbf{V}$ (close to 0) indicates that the group difference explains a large portion of the total variability; therefore, the null hypothesis is not supported, providing strong evidence of a significant difference in group means across the multiple dependent variables. Other test statistics that are computed are the Hotelling-Lawley Trace, the Wilks Lambda, and Roy's Maximum Root.

Dependent variables, also known as response variables, are variables or factors that are assessed and are anticipated to vary as a result of changes in the independent variable(s).

Independent variables are controllable factors manipulated to examine their effects on dependent variables.

A one-way MANOVA will be performed to assess whether the cognitive assessment scores jointly differ across independent groups.

3.4.2 Machine Learning Algorithms

Machine learning algorithms refer to models that allow computers to learn from data and make predictions or decisions without direct instruction. Both supervised and unsupervised learning models will be employed to analyze clinical demographic variables, cognitive assessment scores, and MRI-derived features, to classify subjects into diagnostic categories and predict cognitive deterioration.

3.4.2.1 Supervised Machine Learning Algorithms

Supervised learning is a machine learning method that uses datasets with human-labeled inputs and outputs to train artificial intelligence models. The trained model identifies the key relationships between inputs and outputs, enabling it to predict outputs accurately when presented with new, unlabeled data from real-world situations (Belcic & Stryker, 2024). The selected supervised learning models to be utilized in the study are Gradient Boosting, Support Vector Machine, Random Forest, Neural Network, and Naïve Bayes.

Gradient Boosting

This is an ensemble learning technique that produces accurate predictions by combining multiple decision trees into a single, more accurate model. Developed by Jerome Friedman, this approach uses base models to enhance their strengths, correct errors, and improve overall predictive performance. By uncovering complex patterns within data, gradient boosting proves to be highly effective for various predictive modeling tasks (Clark & Lee, 2025). The Gradient Boosting algorithm integrates multiple viewpoints, which offers a solution to bypass the shortcomings of single models, attain better optimization, and enhance model precision, particularly when working with intricate or noisy datasets.

The model minimizes a regularized objective function, accounting for the empirical loss over training the data:

$$Q(\Theta) = \sum_{i=1}^n L(y_i, F(x_i)) + \sum_{k=1}^K \Omega(f_k), \quad \Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (3.4)$$

- $L(y_i, F(x_i))$ is the loss function, and $\Omega(f_k)$ is the regularization term
- T is the number of leaves in the tree, and w_j is the weight of the j 'th leaf

In this research, we will utilize the Gradient Boosting algorithm due to its effectiveness in managing high-dimensional medical datasets. This is particularly relevant for the OASIS dataset, which includes a variety of MRI-derived features, cognitive evaluation scores, and demographic clinical variables.

Support Vector Machine (SVM)

SVM is a supervised machine learning algorithm used for classification and regression tasks by creating an optimal line, called a hyperplane, that best separates data points into their respective classes in an N-dimensional space. SVM algorithms handle both linear and non-linear classification problems, where kernel functions are used to transform the data into a higher-dimensional space, allowing for separation in cases where non-linear decision boundaries exist.

Here, the model employs a constrained optimization problem to maximize the margin between the hyperplane and the support vectors:

$$\min_{w, b, \xi} \frac{1}{2} \|\vec{w}\|^2 + C \sum_{i=1}^n \xi_i, \quad (3.5)$$

subject to the constraint; $y_i(\vec{w} \cdot \vec{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall_i$

- $\vec{w}$, the weight vector that determines the hyperplane
- C , the regularization parameter that balances the trade-off between margin maximization and minimizing the classification errors
- ξ_i , the non-negative slack variable that permits for misclassification
- b , the bias that maximizes the margin

The SVM algorithm will be employed in this study to classify subjects as demented or non-demented based on the MRI-derived features, cognitive evaluation scores, and demographic clinical variables. This would utilize both linear and non-linear SVMs to identify the optimal hyperplane that maximizes the distance between the different classes of data points.

Random Forest

Random Forest is an ensemble learning technique that operates by utilizing the output of multiple decision trees to develop optimal predictive models, using bagging (bootstrapping aggregating) and feature randomness to enhance model accuracy and stability. The decision trees are constructed using distinct random samples of the dataset to determine the optimal feature splits for subsetting the data. The random forest method, on the other hand, evaluates a random selection of those features, enhancing diversity, maintaining low correlation between the trees, and minimizing the risks of overfitting, bias, and overall variance.

The aggregate prediction function for B trees $\{T_b\}_{b=1}^B$

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (3.6)$$

with T_b , the b'th decision tree in the forest

This study will apply random forest algorithms to develop a forecasting model for Alzheimer's progression, using variables from the OASIS dataset, which include MRI-derived features, cognitive assessment scores, and clinical demographic variables.

Naïve Bayes

This is a supervised stochastic machine learning algorithm that is derived from Bayes' theorem, used for classification problems. The Naïve Bayes algorithm determines the likelihood of a sample from a dataset belonging to a particular class based on the features of the dataset. It relies on the "naïve" assumption of conditional independence and then selects the posterior class with the highest posterior probability from the group of classes.

Bayes' theorem is a mathematical formula that updates the likelihood of a hypothesis or an event given new evidence, permitting the inversion of conditional probabilities. The Bayes' rule is given as;

$$P(Class|Features) \propto P(Features|Class) \times P(Class) \quad (3.7)$$

$P(Class|Features)$ is the posterior probability of a sample belonging to a class given the MRI-derived features, cognitive assessment scores, and clinical demographic variables.

$P(Features|Class)$ is the likelihood of a sample from the dataset having specific MRI-derived features, cognitive assessment scores, and clinical demographic variables.

$P(Class)$ is the prior probability of a sample belonging to a class.

We select the class with the maximum value, the Maximum A posteriori (MAP) rule, the predicted class for an observation;

$$\hat{C}(x) = \arg \max_{C_k} \prod_{j=1}^n P(x_j|C_k) \cdot P(C_k) \quad (3.8)$$

with $\prod_{j=1}^n P(x_j|C_k)$ is the likelihood of the features given the class, and $P(C_k)$ is the prior probability of the $k'th$ class

In this study, the Naïve Bayes algorithm will be applied as a machine learning model to classify subjects into demented and non-demented groups, using the baseline MRI-derived features, cognitive assessment scores, and clinical demographic variables provided by the OASIS dataset.

Neural Network

A Neural Network is a machine learning model that emulates brain-inspired mechanisms by assembling interconnected nodes (known as neurons) in layers. It learns patterns using weights and biases to transform input features into predictions or outputs. Neural networks comprise multiple layers, which are subdivided into an input layer, one or more hidden layers, and an output layer, mimicking the behavior of the human brain. These interconnected nodes function as a linear regression model used for prediction. The varying weights of connections between the nodes determine how each input influences the output layers. Lee (2025) mathematically summarizes neural networks as algorithms that learn a function $f(\mathbf{X})$ by transforming an input vector $\mathbf{X} = (x_1, x_2, x_3, \dots)$ to predict an output $\mathbf{Y}$ in a layered architecture. Information is transmitted from one layer to the next in a Feedforward Neural Network (FNN), indicating a one-way flow of data without any cycles.

The prediction function for a single hidden-layer MLP with K hidden units;

$$\hat{y} = \sigma\left(\sum_{j=1}^K v_j g\left(\sum_{i=1}^n w_{ij} x_i + b_j\right) + c\right) \quad (3.9)$$

$\sigma(\cdot)$, the activation function for the output layer,

$g(\cdot)$, the hidden layer activation function

v_j , the weight linking the j' th hidden unit to the output

w_{ij} , the weight for the i' th input feature connecting the j' th hidden unit

b_j , the bias term for the j' th hidden unit

c , the bias term for the output unit

This study will utilize a simple feedforward neural network to forecast cognitive decline using a layered network architecture based on the OASIS longitudinal dataset.

3.4.2.2 Unsupervised Machine Learning Algorithms

Unsupervised learning, in contrast to supervised learning, is a machine learning method that trains on unlabeled data to find hidden patterns, similarities, and differences without human oversight. Unsupervised learning algorithms operate solely on input data to discover inherent hidden patterns without prior knowledge or training. This research, in conjunction with supervised learning algorithms, will employ several unsupervised learning methods to uncover unseen and novel patterns directly from the OASIS dataset, providing additional insights into the progression of Alzheimer's disease. Principal Component Analysis (PCA) and Clustering will be conducted, utilizing the baseline clinical demographic variables, cognitive assessment scores, and MRI-derived features from the OASIS longitudinal dataset.

Principal Component Analysis (PCA)

PCA is a form of unsupervised learning that reduces the dimensions of a multivariate dataset by transforming correlated variables into fewer, linearly uncorrelated variables called principal components, while minimizing data loss. PCA enables the extraction of the most informative features from large datasets, preserving important statistical information from the original dataset, thereby addressing the issues of multicollinearity and overfitting in regression analysis. By PCA, one seeks to reduce p original variables to k new variables, where k is a compromise between the minimum number of variables and the maximum amount of information retained. The principal components are linear combinations of the original variables in the initial dataset.

We estimate the principal components such that the selected eigenvector maximizes the variance of the estimated principal component.

The optimization model for PCA is;

Estimate $Y_p = \ell_{p1}X_1 + \ell_{p2}X_2 + \dots + \ell_{pk}X_k$ such that $\ell_{p1}, \ell_{p2}, \dots, \ell_{pk}$ maximizes $Var(Y_p)$,

subject to the constraint; $\|\ell_p\|^2 = \ell_p' \ell_p = \sum_{p=1}^k \sum_{l=1}^k \ell_{pl}^2 = 1$

Resulting in;

- $Var(Y_p) = \ell_p' Var(X_i) \ell_p = \ell_p' \Sigma \ell_p = \ell_p' \lambda_p \ell_p = \lambda_p$ (3.10)

- $Cov(Y_p, Y_q) = \ell_p' \Sigma \ell_q = 0, p \neq q = 1, 2, \dots, k$ (3.11)

with λ_p , the eigenvalue (explained variance) of the p' th principal component in the dataset, ℓ_p , is the eigenvector (direction) of the p' th principal component in the dataset, Σ the population variance-covariance matrix of the original variables (X) in the dataset.

Principal Component Analysis will be performed on the OASIS dataset to reduce the dimensionality of the clinical demographic variables, cognitive assessment scores, and MRI-derived features from the OASIS longitudinal dataset. This will allow the principal components to capture the maximum amount of variability in the data.

Clustering

Clustering is an unsupervised learning technique that involves dividing the dataset into groups, called clusters, based on similarities. It does so without using any pre-defined labels or categories to reveal the inherent patterns within the data. By clustering, we assign data points to groups (clusters) based on similarities or the distance between their centroids, with data points in a specific cluster sharing common characteristics. The K-means Clustering algorithm partitions the data into K clusters, assuming that the cluster center defines the Euclidean distance to the centroid, and assigns data points to the nearest centroid through iterative processes (Somarribas & Loteanu, 2020).

The K-means algorithm optimizes the average distance to members within a cluster, modeled as;

$$\sum_{k=1}^K \frac{1}{n_k} \sum_{i=1}^{n_k} \sum_{j=1}^{n_k} \|x_{ki} - x_{kj}\|^2 \quad (3.12)$$

which is the same as minimizing the within-cluster variance defined as;

$$J = \sum_{k=1}^K \sum_{i=1}^{n_k} \|x_{ki} - \mu_k\|^2 \quad (3.13)$$

with μ_k , the k' th cluster mean, x_{ki} , the i' th data point assigned to cluster k , and n_k , the number of data points in cluster k

Including K-means clustering in this study helps identify subgroups of subjects with similar MRI features and cognitive patterns, reducing the complexity of the longitudinal OASIS dataset.

3.5 Model Evaluation Metrics

These tools are crucial for evaluating the effectiveness of predictive models and classification tasks. A variety of performance evaluation metrics will be employed to verify the predictive accuracy of the Alzheimer's disease forecasting model and the classification outcomes, ensuring a comprehensive evaluation. The subsequent metrics will be applied.

1. Accuracy

This measures the overall correctness of the binary classification model, which indicates whether a subject is demented or non-demented, using the baseline MRI and demographic data. The formula is given as;

$$\begin{aligned} \text{Accuracy} &= \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \\ &= \frac{TP + TN}{TP + TN + FP + FN} \end{aligned} \quad (3.14)$$

TP, the true positive, where a demented subject is correctly classified as **demented**

TN, the true negative, where a non-demented subject is correctly classified as **non-demented**

FP, the false negative, where a non-demented subject is incorrectly classified as **demented**

FN, the false negative, where a demented subject is incorrectly classified as **non-demented**

2. Precision (Positive Predictive Value)

This measures the number of correct positive predictions made by the classification model relative to the overall positive predictions, indicating how well the model avoids false alarms and hence the diagnostic reliability of the disease, given as;

$$\textbf{Precision} = \frac{TP}{TP + FP} \quad (3.15)$$

3. Sensitivity

This measures the model's capacity to identify actual dementia by gauging how effectively it recognizes illness when it actually exists

$$\textbf{Sensitivity} = \frac{TP}{TP + FN} \quad (3.16)$$

4. F1-Score

A measure of the harmonic mean of the positive predictive value and sensitivity, showcasing the model's capacity to identify true dementia cases while minimizing false alarms accurately.

$$\textbf{F1 - Score} = 2 \times \frac{\textit{Precision} \times \textit{Sensitivity}}{\textit{Precision} + \textit{Sensitivity}} \quad (3.17)$$

5. Area Under Curve (AUC) and Receiver Operating Characteristic (ROC) Curve

The Receiver Operating Characteristic (ROC) curve simplifies the display of the trade-off between TPR and 1-1-FPR by graphically representing the True Positive Rate (TPR), also referred to as Sensitivity, versus the False Positive Rate, often known as Specificity, of a classification model. It graphically displays the discriminative capacity of the classification model at different probability levels.

$$TPR = \frac{TP}{TP + FN} \quad (3.18)$$

$$FPR = \frac{FP}{FP + TN} \quad (3.19)$$

The Area Under Curve (AUC) is the area enveloped by the ROC curve, indicating how well the classification model distinguishes between the classes. Its value spans 0 to 1, with the subsequent interpretation.

- AUC = 0.5: an indifference between the classification model and random guessing
- AUC = 1: a perfect classification model
- AUC > 0.5: the classifier outperforms random guessing

6. Confusion Matrix

It refers to the tabulated summary of the model's prediction result, examining the difference between predicted and actual classifications to visualize the classification results.

7. Silhouette Coefficient

A clustering metric that integrates cohesion and separation to evaluate how well data points fit within their assigned cluster. Cluster cohesion refers to the similarity of datapoints to their own cluster, while Cluster separation signifies the difference of datapoints in a cluster relative to other clusters. The Silhouette coefficient ranges from -1 to 1, where;

- -1: a wrong assignment of datapoints to a cluster
- 0: a datapoint lies on the decision boundary of two neighboring clusters

- 1: a perfect assignment of datapoints to a cluster

$$\textit{Silhouette Score} = \frac{b - a}{\max(a, b)} \quad (3.20)$$

a = mean distance between intra-cluster datapoints (cluster cohesion)

b = mean distance between a datapoint and all points in the nearest cluster

3.6 Research Ethics

The study exclusively uses a public dataset from the Open Access Series Imaging Studies (OASIS) database. Since no sensitive personal information is included, the study's findings depend on the quality of the OASIS dataset. We obtained this dataset solely for scholarly purposes, contributing to future discoveries in basic and clinical neuroscience.

CHAPTER FOUR

DATA ANALYSIS, RESULTS, AND DISCUSSIONS

4.0 Overview

This chapter presents the results of the data analyses employed to develop the predictive and classification models for Alzheimer’s disease (AD). The analysis integrates both statistical methods and computational techniques to enhance our understanding of the relationships between demographic, cognitive, and MRI-derived factors concerning the onset and progression of Alzheimer’s disease. It evaluates the effectiveness of the selected machine learning models, examines their outcomes, and relates the findings to the study’s objectives. Ultimately, the results are analyzed in connection with previous research findings, highlighting whether the outcomes are consistent with or challenge the current literature on the progression and prediction of Alzheimer’s disease.

4.1 Descriptive Analysis

Descriptive Statistics of the dataset

Descriptive statistics were computed to understand the traits of the dataset and summarize the quantitative variables of the subjects. This included various measures of central tendency, measures of variability, and quantiles, providing insights into the diversity of subjects and the overall structure of the data.

Table 4.1: Descriptive Statistics of Quantitative Variables: A Comprehensive Analysis

Variable	Mean	Median	SD	Q1	Q3	Min	Max
Visit	1.88	2	0.92	1	2	1	5

MR Delay	595.1	552	635.49	0	873	0	2639
Age	77.01	77	7.64	71	82	60	98
EDUC	14.6	15	2.88	12	16	6	23
SES	2.46	2	1.13	2	3	1	5
MMSE	27.34	29	3.68	27	30	4	30
CDR	0.29	0	0.37	0	0.5	0	2
eTIV	1488.13	1470	176.14	1357	1597	1106	2004
nWBV	0.73	0.73	0.04	0.7	0.76	0.64	0.84
ASF	1.19	1.19	0.14	1.1	1.29	0.88	1.59

Table 4.1 presents the summary statistics of the dataset's key features. The dataset included a substantial number of older adults, as indicated by a mean age of approximately 77 years, with most participants falling between 71 (Q1) and 82 (Q3) years old. This relates to the typical age range associated with Alzheimer's disease, with a sizeable portion of the sample exhibiting, on average, mild cognitive decline (mean MMSE score of 27.34). This suggests that the dataset represents an older population, comprising both individuals with dementia and those without. This age range aligns with the population most affected by Alzheimer's disease and mild cognitive impairment.

Visit numbers range from 1 to 5, with a mean of 1.88, indicating that most subjects had between one and two follow-up sessions. This confirms the longitudinal nature of the dataset but also suggests that repeated measures are relatively limited for many participants. MR Delay, which measures the time in days since the baseline scan, has a wide range (0–2639 days) and a high

standard deviation of 635.49. This shows considerable variability in follow-up intervals, a common feature of real-world longitudinal clinical data.

Education (EDUC) averages 14.6 years (SD = 2.88), indicating that the sample generally has relatively high educational attainment, a factor known to influence cognitive reserve and the progression of dementia. Socioeconomic status (SES) exhibits moderate variability (mean = 2.46, SD = 1.13), with values spanning 1 to 5. This indicates a diverse sample in terms of social background, supporting generalizability.

MMSE scores range from 4 to 30, with a mean of 27.34 (SD = 3.68). Since the maximum MMSE score is 30, these values suggest that most participants are cognitively normal or mildly impaired. However, the lower tail (minimum = 4) confirms the presence of more advanced impairment cases. The Clinical Dementia Rating (CDR) has a mean of 0.29 (SD = 0.37), with scores from 0 to 2. The median of 0 indicates that most participants are nondemented at baseline, while the small number of higher scores (1–2) includes individuals with mild to moderate dementia.

In terms of MRI-derived variables, eTIV (estimated total intracranial volume) exhibits significant variation (mean = 1488, SD = 176), reflecting the natural diversity in head sizes. The nWBV (normalized whole brain volume), with an average of 0.73 (SD = 0.04), suggests that participants preserve approximately 73% of their normalized brain volume, a finding that aligns with older adults, including those exhibiting early signs of neurodegeneration

4.2 Exploratory Data Analysis Results

4.2.1 Distribution of Dementia Status

The dataset displays that non-demented cases are predominant among the subjects. At the same time, a lesser percentage experienced dementia, and an even smaller group underwent transition from a non-demented condition to dementia during the study's duration. This expected imbalance is indicative of the dataset's inclusion of a substantial number of normal aging adults, along with Alzheimer's disease-diagnosed subjects.

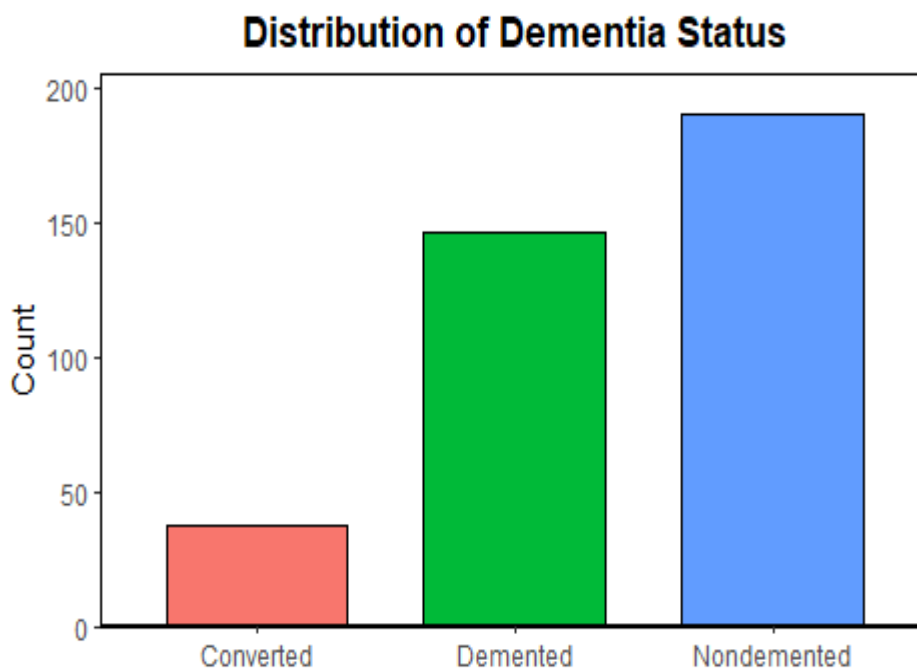


Fig. 4.1: Bar chart of dementia status

4.2.2 Gender and Dementia Status Distribution: A Bivariate Analysis

A bivariate analysis of the dataset, considering the gender composition and dementia status of each subject, revealed a higher prevalence of dementia among males, despite the larger proportion of females in the dataset. This pattern is inconsistent with the literature, as women

are more likely to develop dementia, chiefly driven by their higher survival rate of old age, biological differences, and gendered social roles and responsibilities (Beam et al., 2018; Menzel et al., 2019; Andrew & Tierney, 2018).

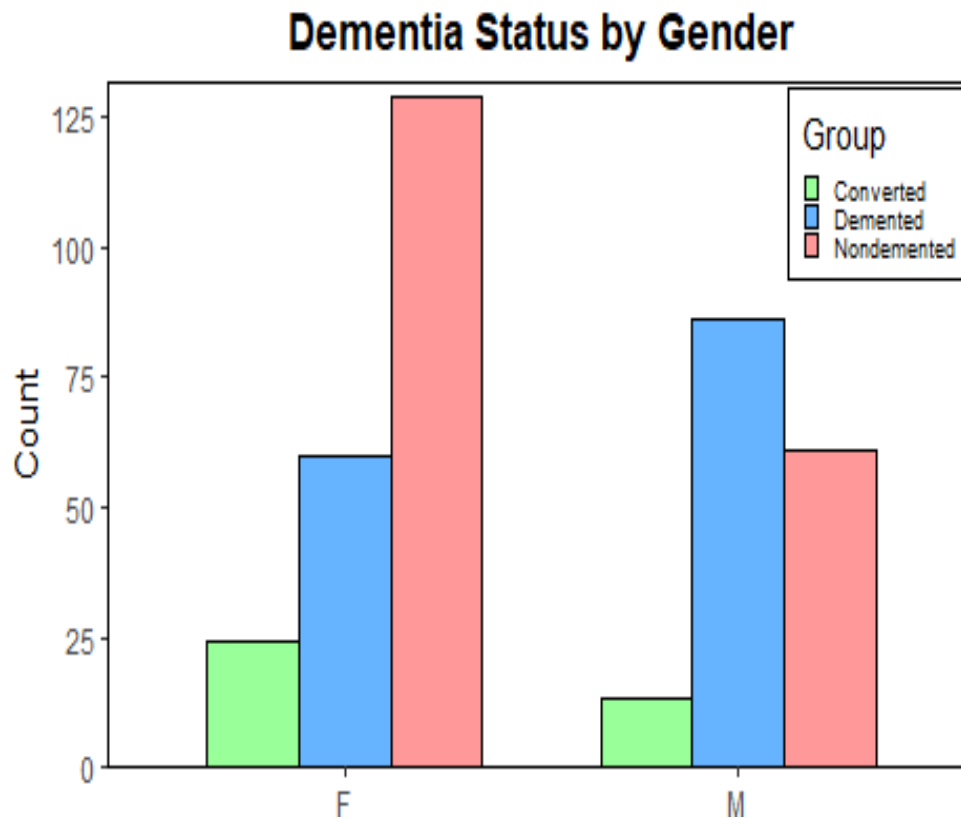


Fig. 4.2: Bar chart illustrating gender differences in dementia status

4.2.3 Age Disparities and Dementia Status: A Bivariate Analysis

The age group analysis of the dementia status discovered that demented subjects were, on average, connected to advancing age, with age groups 71-80 and 81-90 displaying a higher frequency of demented cases. A similar observation is made about the converted subjects: the conversion rate increases with age, peaking in the 81-90 age group, followed by the 71-80 age group. This trend fortifies the well-established link in the literature and real-world situations between increasing age and the risk of developing Alzheimer's disease. Quite interesting,

fewer subjects were classified as demented and converted in the age group 91-100, indicative of other potential demographics in the dataset that might affect the dementia status.

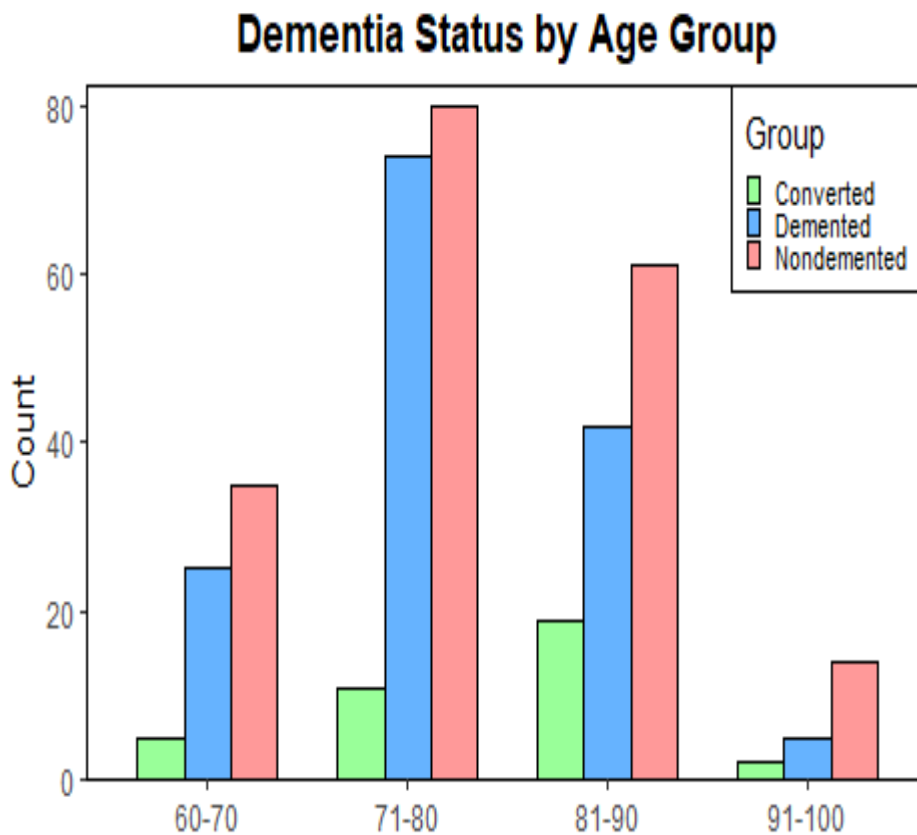


Fig. 4.3: Bar chart of age group distributions by dementia status

4.2.4 Bivariate Analysis of Cognitive Assessment Tools and Dementia Status

A box plot analysis of the cognitive evaluation of tools (Mini-Mental State Examination scores and Clinical Dementia Rating scores) and the dementia classification depicts a coherent and uniform association.

Figure 4.4 shows the Mini-Mental State Examination (MMSE) scores over the dementia status categories. The analysis reveals that the MMSE scores were higher among non-demented subjects, demonstrating normal cognition, followed by subjects who transitioned from a

healthy state to a demented state. Subjects classified as demented, however, showed considerably lower MMSE scores, signaling a notable decline in cognitive abilities. This trend is expected and consistent with public health research and global trends.

The Clinical Dementia Rating (CDR) box plot analysis in Figure 4.5 displays lower CDR scores for non-demented participants, with an approximate CDR score of zero. Converted subjects exhibited moderate CDR scores, ranging from 0 to 0.5, indicating a gradual progression in cognitive decline over the study period. CDR scores were relatively high in dementia classified subjects (CDR scores spanned 0.5 to 2), indicative of a decline in neurodegenerative conditions and deterioration in functionality.

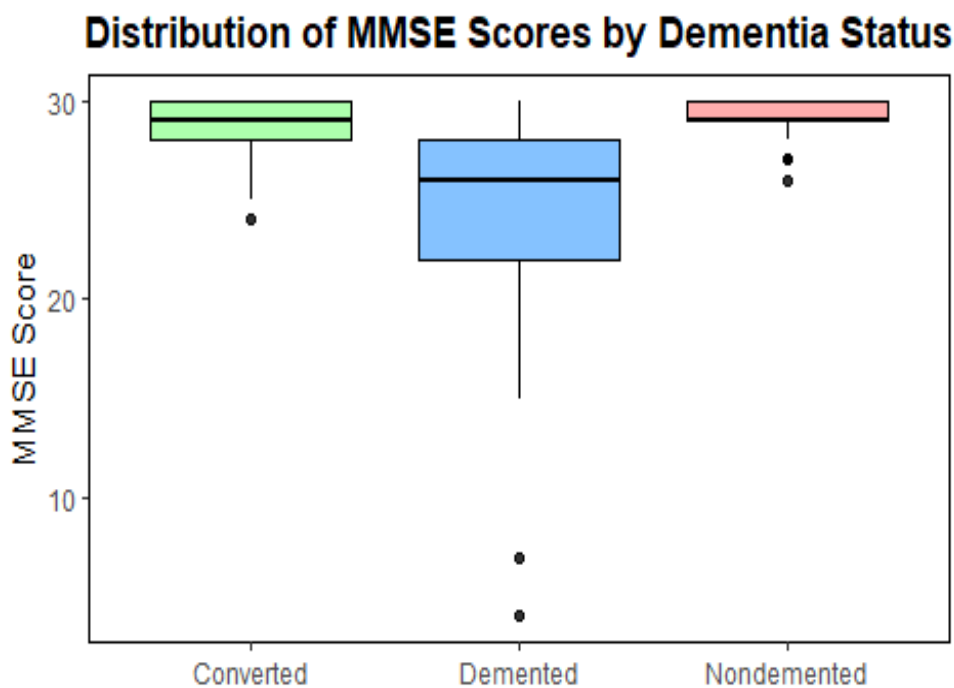


Fig. 4.4: Box plot showing the distribution of MMSE scores across dementia statuses

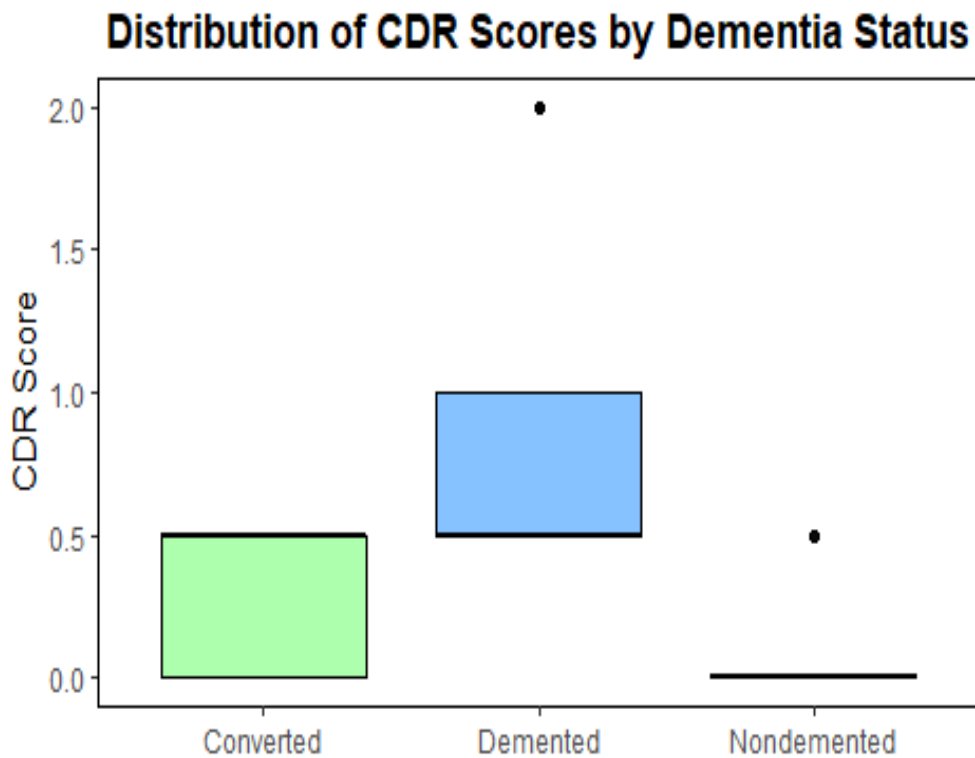


Fig. 4.5: Box plot showing the distribution of CDR scores across dementia statuses.

4.2.5 Dementia Status by Socioeconomic Disposition: A Bivariate Analysis

The socioeconomic disposition of subjects clearly and somewhat oppositely influenced dementia status in the dataset. Participants with elevated socioeconomic scores showed significantly greater levels of dementia; those rated four in economic and social standing were most affected, followed by those rated three. Non-demented individuals were more common in lower socioeconomic categories, with grade two subjects showing the highest proportion of non-demented cases. This pattern conflicts with existing research and global neuroscience data, which indicate that individuals with lower socioeconomic status (SES) are more likely to develop dementia. Subjects who shifted from healthy cognition to progressive neurodegeneration during the study demonstrated an inverse relationship with high socioeconomic status ratings.

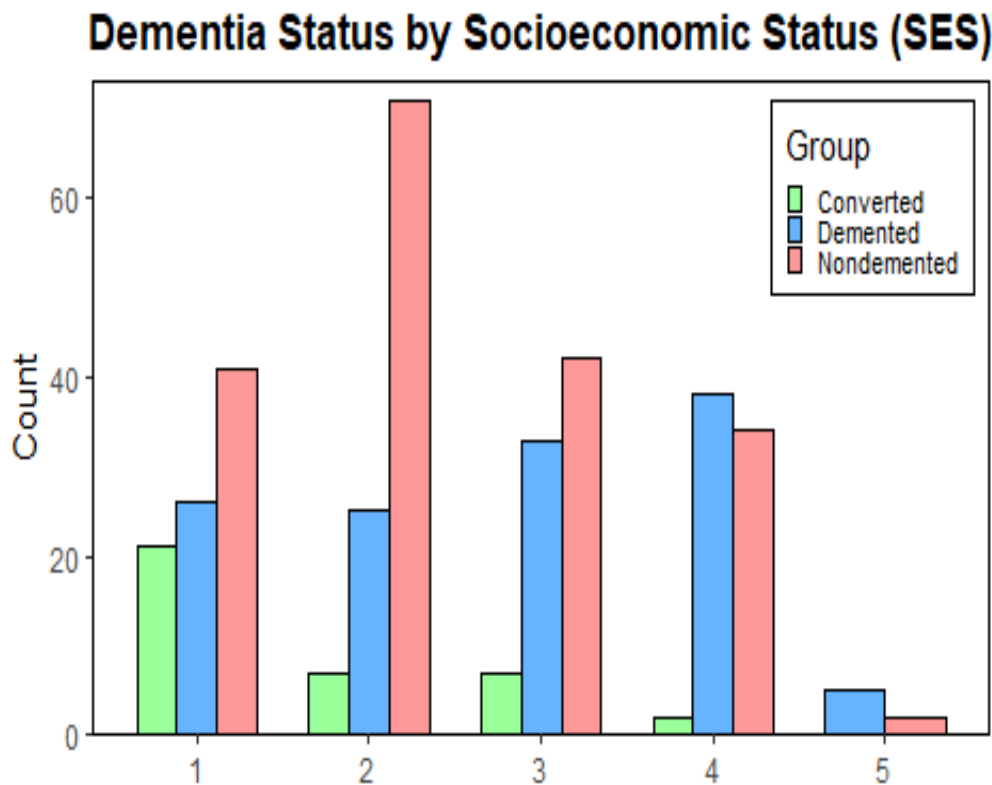


Fig. 4.6: Bar chart of dementia status by socioeconomic disposition

4.2.6 Analysis of Dementia Status Based on Education: A Bivariate Examination

The influence of education level on dementia classification in the dataset is highly significant. A bivariate bar chart analysis of dementia categories and educational levels revealed that higher education levels are associated with a lower prevalence of dementia among the subjects. Participants with the highest level of education (21 to 23 years of schooling) had the fewest cases of dementia, while non-demented individuals were more common at higher education levels. This trend aligns with established research and worldwide observations, as higher educational attainment generally provides a greater safeguard against cognitive impairments, a concept clearly articulated by the Cognitive Reserve Theory (Stern, 2007).

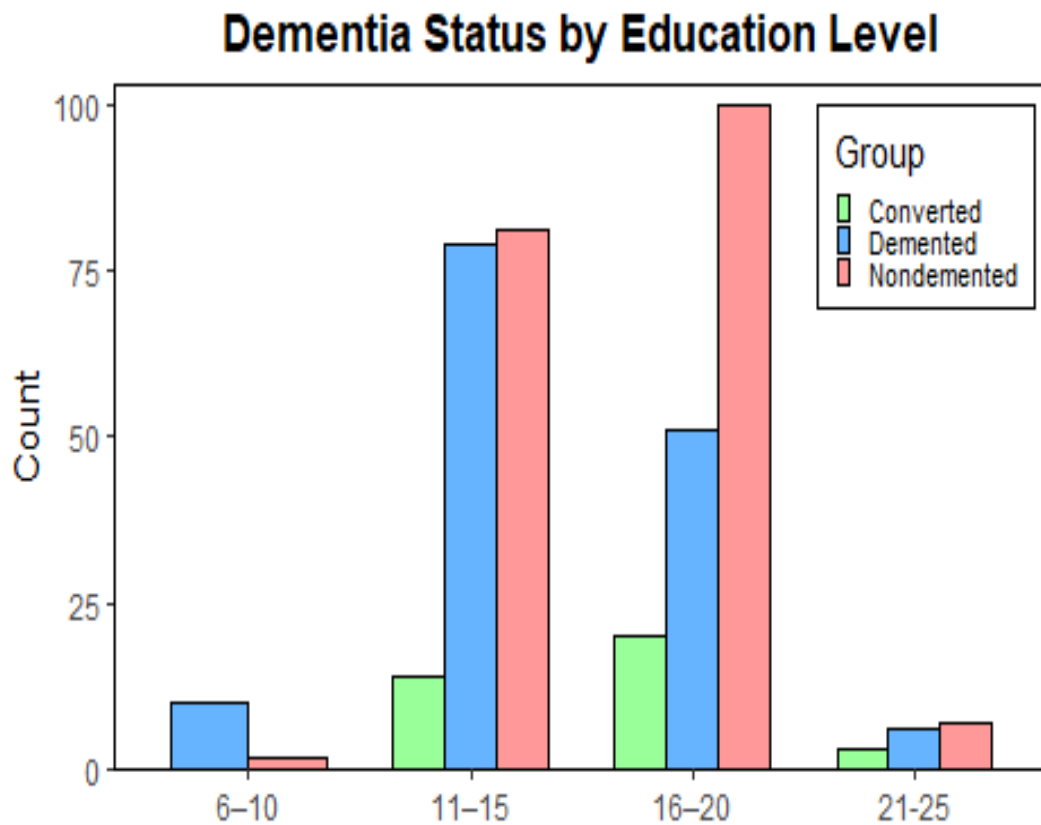


Fig. 4.7: Bar chart illustrating dementia statuses across education levels.

4.2.7 MRI-derived Features and Cognitive Status

An analysis using a bivariate box plot of the normalized Whole Brain Volumes (nWBV), categorized by dementia status, revealed a consistent and uniform relationship between the two. Normalized whole-brain volumes were relatively higher among non-demented subjects, indicating a greater amount of intact brain tissue. In comparison, participants classified as demented exhibited noticeably lower normalized whole-brain volumes. This reflects the measurable shrinkage of notable brain regions. Subjects who metamorphosed from a healthy condition to one of dementia exhibited moderate nWBV values, indicating a progressive reduction in brain volume. This trend is expected and aligns with existing literature; the

Atrophy-Cognition Model, which posits progressive neurodegeneration in distinct brain regions of an AD patient, corresponds to a gradual decline in cognition.

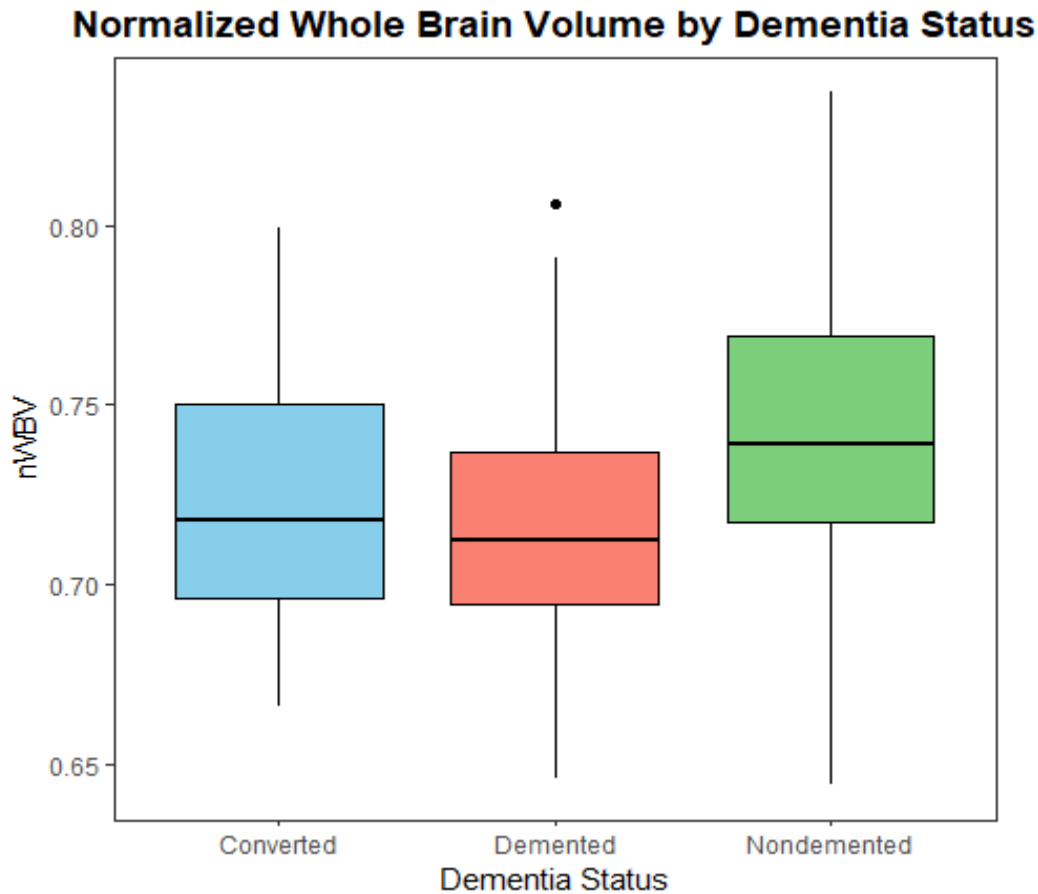


Fig. 4.8: Box plot displaying the nWBV values categorized by dementia statuses.

4.2.8 Correlation Analysis and Multivariate Relationships

A Pearson correlation analysis, used to identify multivariate relationships and spot patterns within the dataset, uncovered the following key insights.

- The Mini-Mental State Examination (MMSE) showed a significant negative correlation with the severity of dementia, as measured by the Clinical Dementia Rating (CDR);

$r \approx -0.71$. This finding aligns with clinical expectations, as more advanced stages of dementia generally lead to lower scores on the MMSE.

- Normalized whole brain volume revealed a mild negative correlation with dementia severity (CDR); ($r \approx -0.35$). This suggests that individuals with smaller nWBV values are more likely to have high dementia severity.
- Age and normalized whole brain volume (nWBV) exhibited an intermediate negative correlation ($r \approx -0.53$). Older adults generally have smaller normalized whole brain volumes.
- Atlas Scaling Factor (ASF) correlates almost negative perfectly with the Estimated Total Intracranial Volume (eTIV); $r \approx -0.99$. This demonstrates a mathematical relationship, where the eTIV is calculated by dividing a constant by the ASF.
- Dementia severity, as assessed by CDR, tends to have a very weak individual correlation with variables such as socioeconomic status (SES), atlas scaling factor (ASF), age, education (EDUC), and estimated Total Intracranial Volume (eTIV). They, however, establish a robust statistical basis for the predictive modeling and MANOVA analysis, guaranteeing that all relationships are valid when combined with other predictors

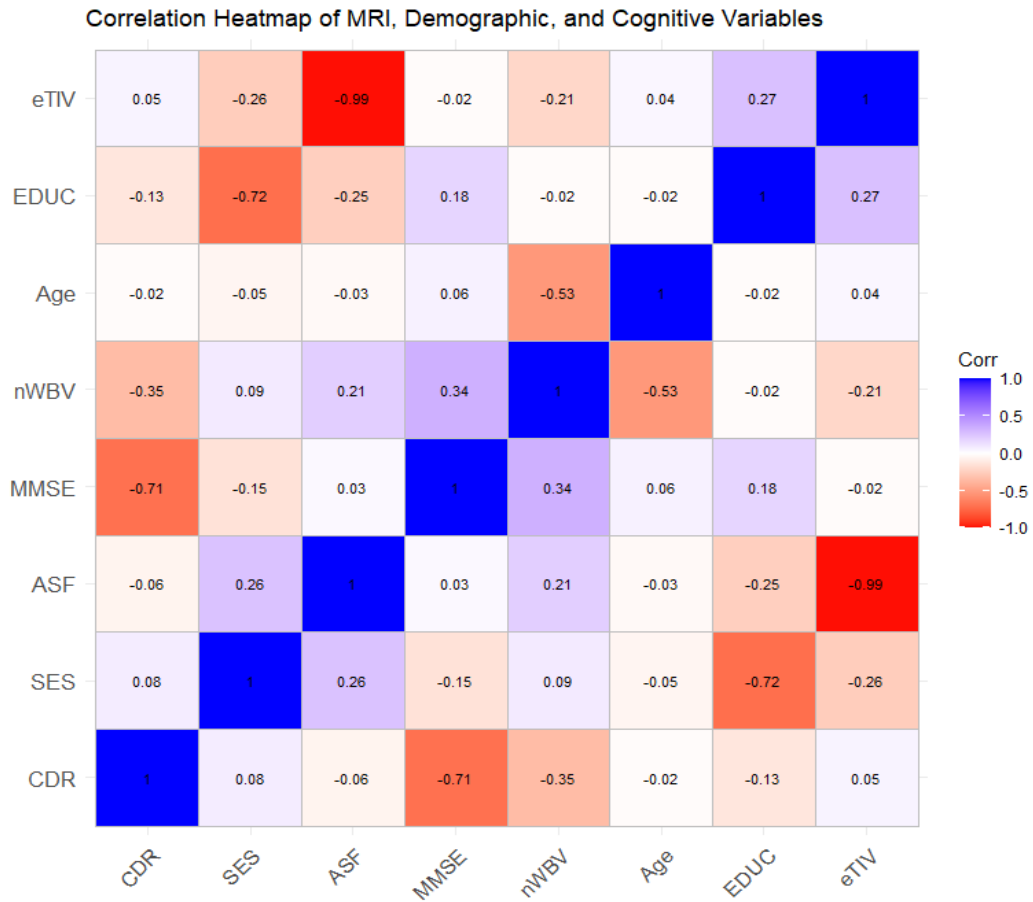


Fig. 4.9: Correlation heatmap illustrating the relationship between variables

4.3 Multivariate Analysis of Variance (MANOVA) Results

A multivariate analysis of variance (MANOVA) was conducted to investigate whether there was a significant difference in the cognitive scores and MRI-derived features across the dementia status, as measured by the *Group*. A preliminary assumption testing conducted revealed the following key insight;

- **Multicollinearity:** The correlation coefficients were less than 0.9; thus, the assumption of multicollinearity is tenable.

- **Independence:** A one-way repeated measures MANOVA was conducted to suit the non-independent observations, given the longitudinal structure of the dataset.
- **Normality:** The Shapiro-Wilk test for the dependent variables showed a breach of the normality requirement ($p < 0.01$). Nevertheless, due to the large sample size in each group, the robustness of MANOVA is adequate to account for this violation, in line with the Central Limit Theorem.
- **Outliers:** While boxplot analysis on the variables indicated potential univariate outliers, all cases were retained since all the values fall within the clinically standard range of the variables. The presence of a few multivariate outliers was also detected using the Mahalanobis distance (a total of 2 cases at a one-percent significance level). Considering the large sample size and the diversity in clinical characteristics, all cases were included in the analysis.
- **Linearity:** The assumption of linearity is satisfactory per the inspection of the scatterplot matrix.
- **Variance-covariance homogeneity:** The Box's M test did not meet the requirement of a homogeneity of the variance-covariance matrix ($p < 0.01$). Consequently, Pillai's Trace is the preferred test statistic to employ, as it has demonstrated greater resilience to these breached assumptions.

Table 4.2: Repeated Measures MANOVA Results Using Pillai's Trace

Effect	Pillai's Trace	df (num, den)	F	p-value
Group	0.446	6, 694	33.240	< 0.001
Visit	0.054	3, 346	6.533	0.00026
Group $\times$ Visit	0.081	6, 694	4.883	< 0.001

The outcome of the RM-MANOVA revealed a statistically significant multivariate difference among the groups (demented, non-demented, and converted) on the combined dependent variables (MMSE, eTIV, and nWBV), $\text{Pillai} = 0.446, F(6, 694) = 33.240, p < 0.001$. The null hypothesis was therefore rejected, leading to the conclusion that dementia status correlates with observable variations in brain structure and cognitive abilities. A follow-up univariate ANOVA was conducted at a five-percent alpha level.

The result indicated statistically sufficient evidence to reject the null hypothesis that the MMSE scores and nWBV values do not differ across the dementia status, $F(2, 356) = 130.658, p < 0.001, \text{partial } \eta^2 = 0.42$, and $F(2, 356) = 21.381, p < 0.001, \text{partial } \eta^2 = 0.11$, respectively, concluding that the dementia status is related to MMSE scores and nWBV values. The effect size was large in both variables. Outcome of the univariate ANOVA on eTIV revealed that eTIV values did not differ across dementia status with a small effect size $F(2, 356) = 0.657, p = 0.519, \text{partial } \eta^2 = 0.004$.

4.4 Principal Component Analysis (PCA) Results

A principal component analysis was conducted on the numeric variables (Age, MMSE, nWBV, eTIV, Visit, MR Delay, EDUC, and SES) to decrease the dimensionality by uncovering hidden patterns in the dataset and converting correlated variables into a smaller set of linearly uncorrelated variables, all while preserving data integrity. Before conducting the analysis, the dataset was normalized to guarantee that all variables were on a consistent scale.

The analysis's results showed that the initial five principal components account for a significant portion of the variation (**91.7%**), leading to their retention for further examination and interpretation. Component 1 explains (**27.68%**) of the variation, component 2 explains

(**23.13%**), component 3 explains (**18.5%**), component 4 explains (**12.79%**), and component 5 accounts for (**9.61%**) of the variation.

The final three principal components account for negligible information (**8.3%**), and hence their rejection.

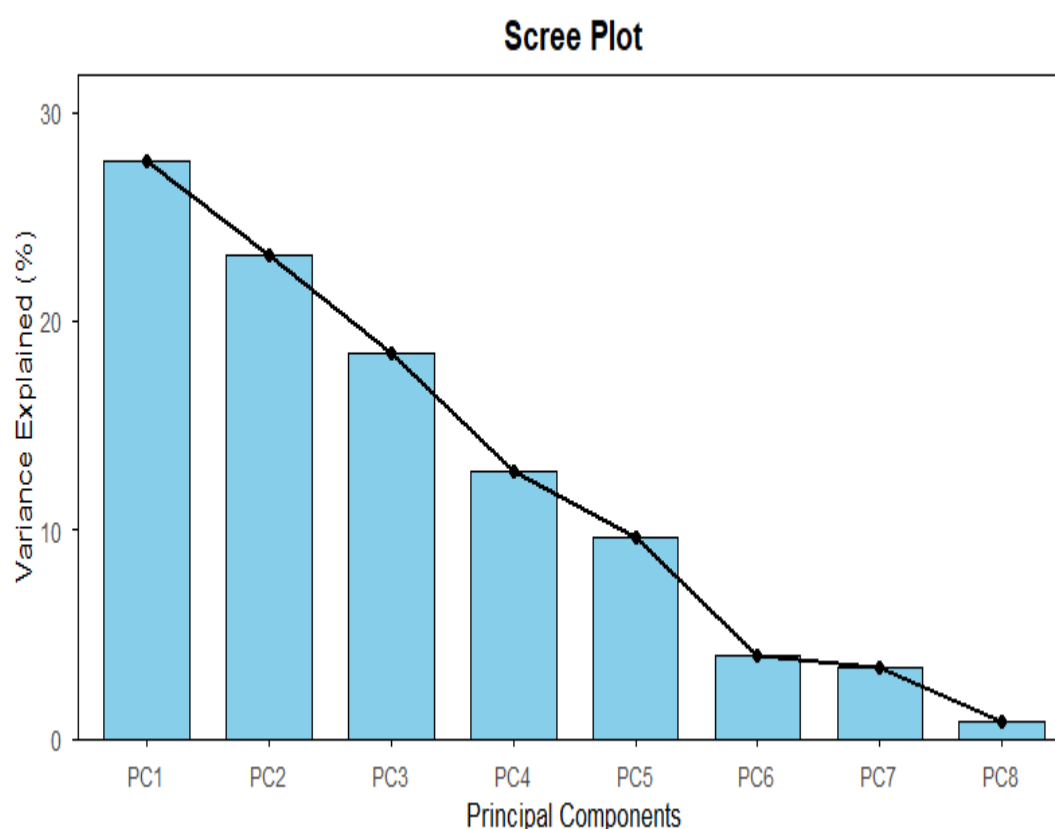


Fig. 4.10: A scree plot illustrating the principal components along with the percentage of variance accounted for.

Analysis of the component loadings revealed that the initial component primarily captures visit number and the time (delay) between different MRI scans in a positive light. The second component positively correlates with socioeconomic disposition, while negatively correlating with educational gradient. PC3 primarily signifies the negative impact of brain volume integrity

and cognitive score, whereas PC4 appears to be negatively associated with cognitive function. PC5 shows a negative correlation with intracranial volume. The variable loadings for the initial four principal components are shown in the table below.

Table 4.3: Component Loadings

Variables	Components				
	PC1	PC2	PC3	PC4	PC5
Visit	0.528	0.281	-0.339	0.148	-0.105
MR Delay	0.528	0.264	-0.377	0.064	-0.058
Age	0.32	0.185	0.368	-0.626	0.056
EDUC	0.267	-0.592	0.009	0.031	-0.264
SES	-0.296	0.565	-0.073	0.002	0.308
MMSE	0.02	-0.263	-0.422	-0.654	0.391
eTIV	0.293	-0.226	0.178	0.386	0.814
nWBV	-0.307	-0.164	-0.627	0.063	0.053

A 3D representation of the PCA outcomes based on dementia status provided a visual overview of both demented and non-demented individuals across the first three principal components. The 3D PCA visualization (as shown in Fig. 4.11) revealed a mild distinction between participants with dementia and healthy subjects, with significant overlap observed. For clarity purposes and easy interpretability, subjects who transitioned from a healthy state to a demented condition were excluded. This reinforces the argument that the retained components extract significant information from the dataset in a reduced dimension.

3D PCA Visualization by Group

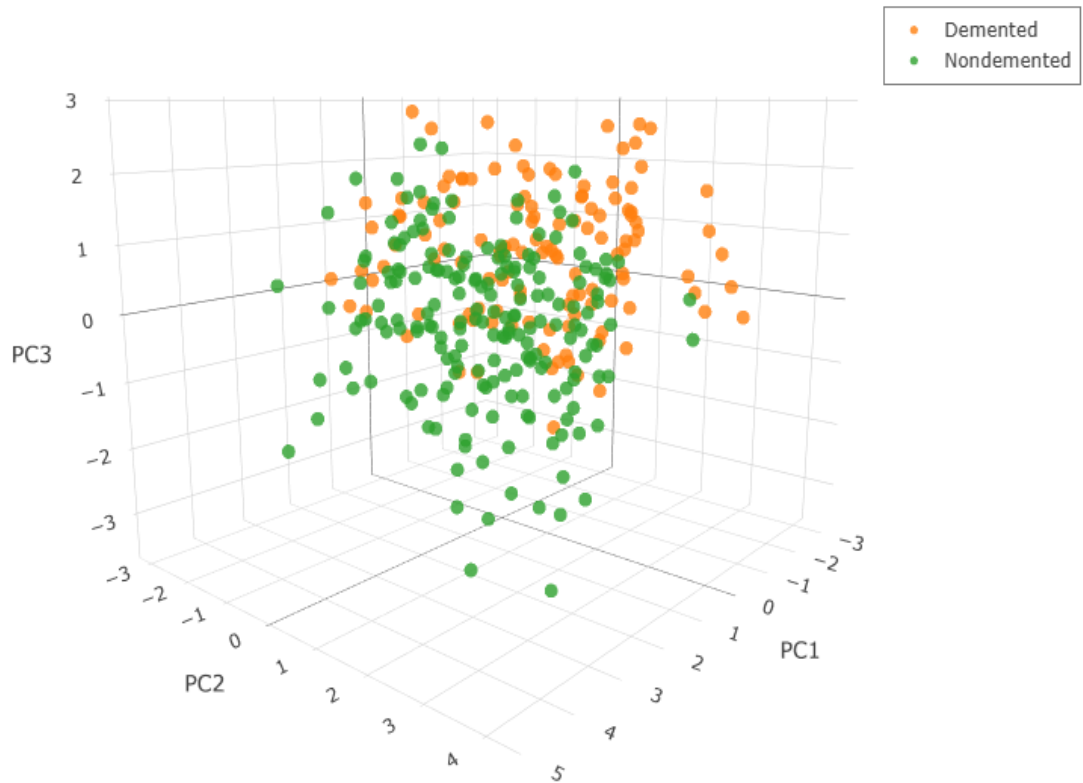


Fig. 4.11: 3D Visualization of principal components by dementia status

4.5 K-Means Clustering Utilizing Principal Components

A K-means clustering analysis was performed to categorize the observations into similar groups, following the dimensionality reduction through Principal Components Analysis. This analysis indicated that the first five components effectively capture a significant amount of information in the dataset. The optimal number of clusters (k) was identified using both the Elbow method and the Silhouette score. The Elbow method, as analyzed by the Within-Sum-of-Squares (WSS) plot, indicated that the optimal number of clusters is four ($k = 4$), implying that the retained PCA-transformed data can be divided into four distinct, homogeneous

subgroups. The Silhouette score analysis revealed an optimal number of clusters to be four, ($k = 5$), suggesting that a solution with four clusters offers the most transparent and most reliable clustering framework for your data. Consequently, five was selected as the ideal number of clusters, supported by the Silhouette analysis because of its recognized reliability and statistical validity.

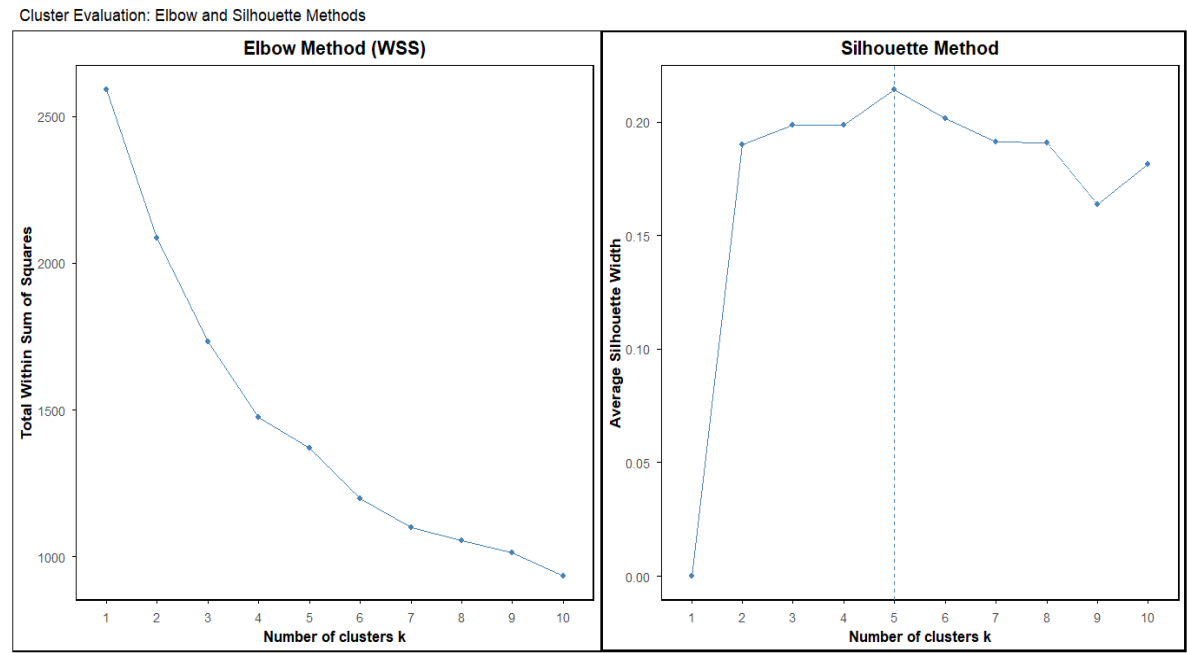


Fig. 4.12: Comparative cluster evaluation

A two-dimensional graphic representation of the K-means clustering (Fig. 4.14), based on the first four principal components and $k = 5$, revealed five clearly defined groups based on similarities in brain structure, cognitive characteristics, and clinical demographics, with significant overlaps. The clustering effectiveness was assessed using the silhouette coefficient, yielding an average silhouette score of 0.214, indicating a weak but discernible clustering structure, which is consistent with the clinical dataset in the existing literature.

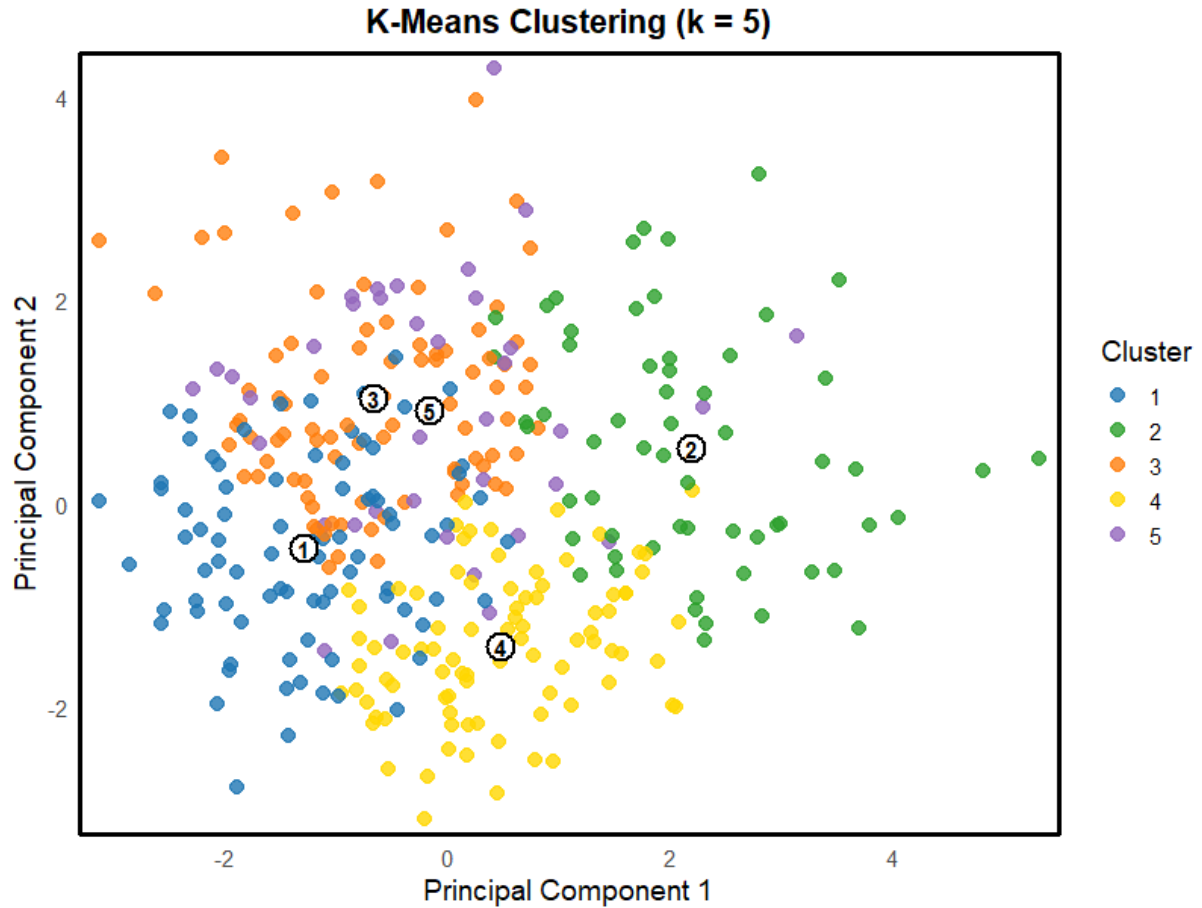


Fig. 4.13: K-means clustering with five clusters based on the first five principal components.

4.6 Developing Models to Forecast and Classify Dementia Progression

This section outlines the result of the predictive modeling phase, concentrating on the effectiveness of the five machine learning algorithms: Gradient Boosting, Support Vector Machine (SVM), Random Forest, Neural Network, and Naïve Bayes. We aim to assess how well these models predict dementia progression, thereby determining which approach best captures the relationship between the response variable (Clinical Dementia Rating) and the predictors (retained principal components and gender). It is worth noting that the Synthetic

Minority Over-sampling Technique (SMOTE) was employed to address the high-class imbalance in the dataset.

Comparison of Model Performance

Table 4.4 offers a comparative overview of the effectiveness of all models used in developing the predictive model, highlighting their test accuracies and cross-validation scores.

Table 4.3: Summary Comparison of the Performance Across All Predictive Models

Model	Test Accuracy	CV Accuracy (±SD)	CV F1 Macro (±SD)
Gradient Boosting	74%	75.3% (±4.2%)	74.9% (±13.5%)
Random Forest	76%	73.6% (±5.4%)	69% (±21.5%)
SVM	72.1%	75% (±5.6%)	73.9% (±15.4%)
Neural Network	75%	75% (±4.4%)	70.8% (±21.1%)
Naïve Bayes	74%	70.9% (±4.6%)	71.5% (±19.9%)

4.6.1 Gradient Boosting

The Gradient Boosting model demonstrated superior performance on the test dataset. Since the cross-validation results are roughly comparable to the test accuracy, this indicates the model's stability and suggests it performs pretty well on new data.

- **Test Accuracy:** 74%
- **Cross-Validation Accuracy:** 75.3% ($\pm 4.2\%$)
- **Cross-Validation F1 (Macro):** 74.9% ($\pm 13.5\%$)

However, careful tuning is necessary to attain optimal performance.

4.6.2 Random Forest

The Random Forest algorithm demonstrated the best performance on both the cross-validation and test datasets, reflecting its effectiveness in practical applications. The significant variability between the F1 score and accuracy within the cross-validation dataset suggests that the model is inconsistent and sensitive to the trained set.

- **Test Accuracy:** 76%
- **Cross-Validation Accuracy:** 73.6% ($\pm 5.4\%$)
- **Cross-Validation F1 (Macro):** 69% ($\pm 21.5\%$)

Despite considerable generalization, the model still struggled with cases of high-severity dementia, suggesting that class imbalance remained a problem, even after applying SMOTE resampling.

4.6.3 Support Vector Machine (SVM)

The SVM classifier demonstrated a moderate level of performance, albeit comparatively lower, on the test set. The model exhibits a good balance between precision and recall, as demonstrated by its consistent performance in cross-validation.

- **Test Accuracy:** 72.1%
- **Cross-Validation Accuracy:** 75% ($\pm 5.6\%$)
- **Cross-Validation F1 (Macro):** 73.9% ($\pm 15.4\%$)

Further fine-tuning the model's parameters could enhance its performance.

4.6.4 Neural Network

The Neural Network algorithm demonstrated a competitive level of accuracy in testing, with perfect consistency between the test set and the validation set. The Neural Network classifier achieved a comparatively lower cross-validation F1 score, which reflects its inconsistency in balancing false positives and false negatives within the severe dementia category.

- **Test Accuracy:** 75%
- **Cross-Validation Accuracy:** 75% ($\pm 4.4\%$)
- **Cross-Validation F1 (Macro):** 70.8% ($\pm 21.1\%$)

4.6.5 Naïve Bayes

The Naïve Bayes algorithm yielded a moderate improvement in test results, with an acceptable level of consistency in accuracy across both the test and validation sets.

- **Test Accuracy:** 74.0%
- **Cross-Validation Accuracy:** 70.9% ($\pm 4.6\%$)

- **Cross-Validation F1 (Macro):** 71.5% ($\pm 19.0\%$)

4.6.6 Comparative Analysis

- The Random Forest model achieved the highest overall accuracy on the test set at 76.0%, outperforming the Neural Network, which reached a test accuracy of 75.0%, as well as Naïve Bayes and Gradient Boosting at 74.0%. SVM achieved the lowest accuracy at 72.1%
- Although Random Forest demonstrated the highest effectiveness overall, all models continued to face difficulties with the severely demented category, emphasizing that class imbalance remained a significant issue despite the application of SMOTE resampling.
- In the process of cross-validation, Gradient Boosting achieved the highest accuracy at 75.3%, with SVM and Neural Network not far behind at 75.0%. Random Forest followed with an accuracy of 73.6%, and Naïve Bayes came in at 66.7%.

4.7 Model Comparison Metrics

Table 4.5 below presents a comparative summary of the performance across all models. The Gradient Boosting and Random Forest models exhibited outstanding performance on the merged dataset, whereas the SVM, Neural Network, and Naïve Bayes models attained a moderate level of overall performance.

Table 4.5: Model Comparison Metrics

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
Gradient Boosting	0.926	0.913	0.919	0.911
Random Forest	0.931	0.908	0.923	0.912
SVM	0.774	0.752	0.703	0.714
Neural Network	0.777	0.773	0.738	0.729
Naïve Bayes	0.736	0.743	0.689	0.705

4.7.1 Receiver Operating Characteristic–Area Under the Curve (ROC–AUC) Analysis

The models' predictive performances were assessed using the Receiver Operating Characteristic–Area Under the Curve (ROC–AUC) metric, which provides a more comprehensive evaluation than just accuracy. The ROC–AUC curve illustrates the connection between each model's True Positive Rate (TPR) and False Positive Rate (FPR) across all decision thresholds. To encapsulate the results across the four classifications of dementia severity, a micro-average was computed. The diagonal dashed line symbolizes random guessing and serves as a baseline, with an AUC of 0.50.

The ROC-AUC assessment revealed an outstanding differentiation of dementia severity, indicated by AUC values exceeding 0.85 across all models. Neural Networks demonstrated the strongest discriminative power, with an AUC of 0.919, while Random Forest followed with an

AUC of 0.888. SVM and Naïve Bayes were not far behind, achieving AUCs of 0.884 each. Gradient Boosting produced a moderate AUC of 0.863.

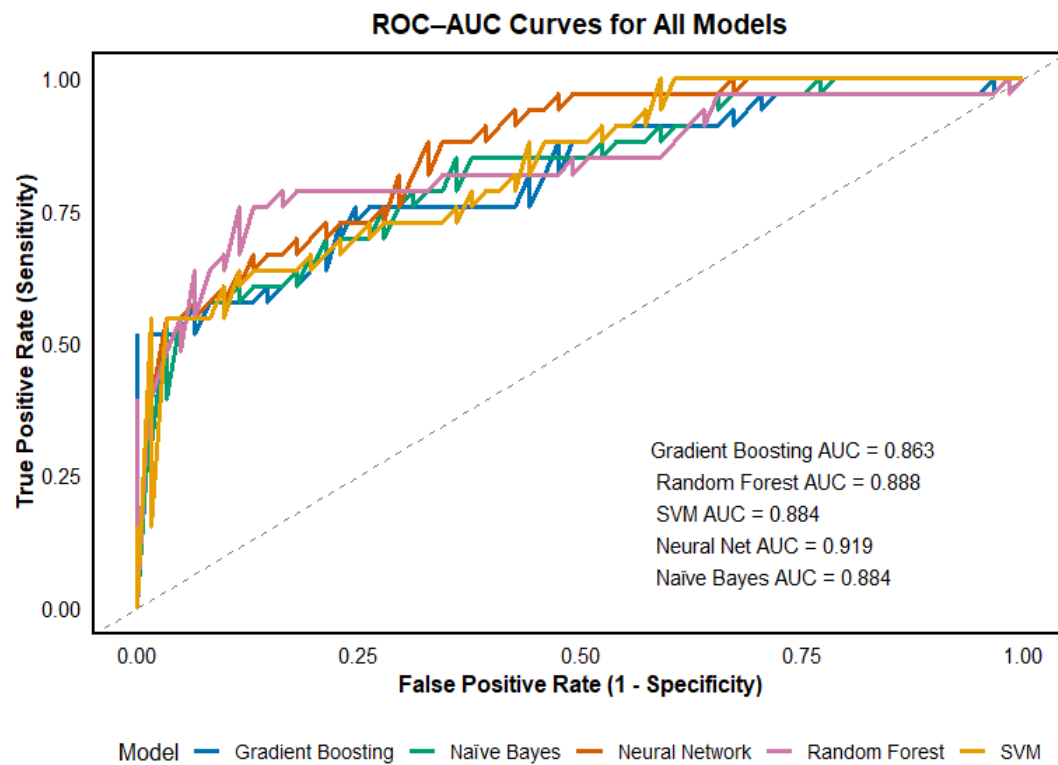


Fig. 4.14: ROC-AUC analysis

4.7.2 Confusion Matrices

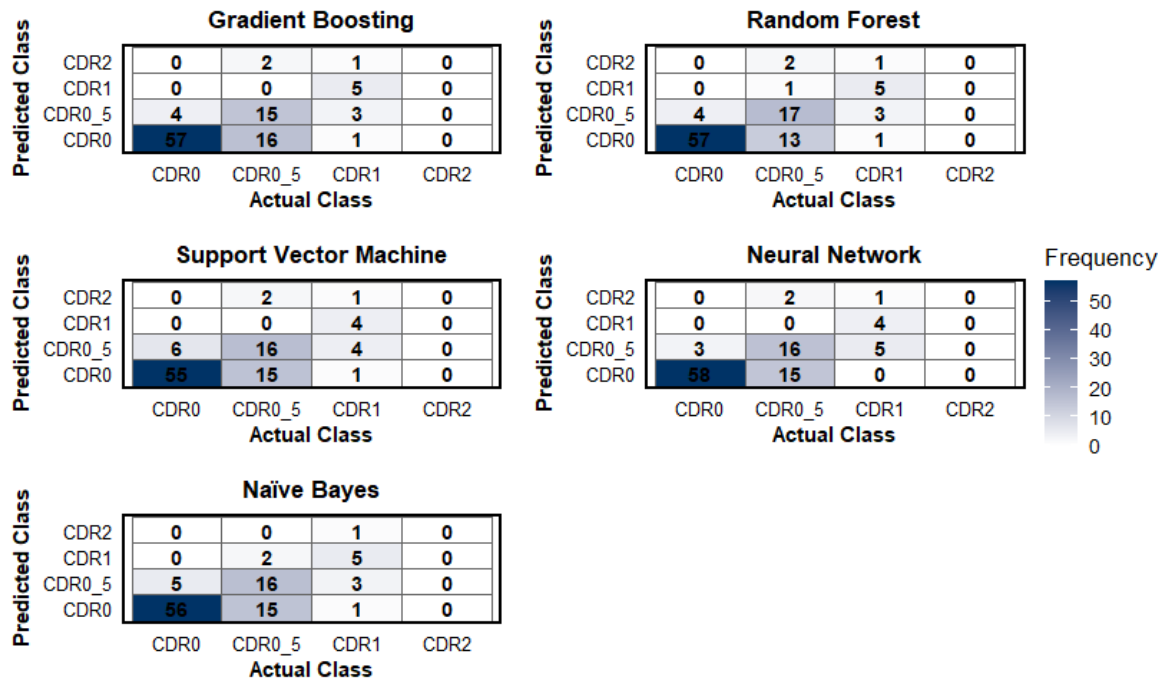


Fig. 4.15: Combined confusion matrices for all the models

Interpretations:

Due to the significant class imbalance in the dataset, even with SMOTE resampling, all models struggled to accurately predict and classify mild and severe dementia cases, resulting in a considerable number of misclassifications in the CDR 0.5 and CDR 1 classes. It is worth noting that the CDR 2 class was not represented in our test set.

- Neural Networks accurately identified most of the non-demented individuals (CDR0), successfully classifying 58 out of 61 participants. This was supported by the highest AUC score of 0.919.
- Random Forest demonstrated outstanding performance, accurately classifying 17 of the 33 very mild dementia cases and achieving an AUC score of 0.888.

- Naïve Bayes, Gradient Boosting, and Random Forest exhibited a competitive performance in classifying moderate dementia (CDR 1) cases, correctly predicting 5 out of 10 cases.
- Support Vector Machine, just like other models, exhibited a similar pattern of misclassification in the CDR 0.5 and CDR 1 classes.

4.8 Model Selection

The Random Forest model was identified as the best classifier according to the evaluation metrics mentioned earlier, attaining an overall accuracy of 93.1% on the integrated dataset. This demonstrates the model's exceptional performance. The significance of each predictor variable in the chosen Random Forest model was evaluated using the Gini index. The Gini index measures the extent to which a variable reduces impurity in a decision tree, indicating the significance of each predictor within the model.

The analysis revealed the third principal component as the most important predictor, with a 42.9% mean decrease in the Gini index. The second principal component also exhibited excellent predictor performance, with a 31.6% mean decrease in the Gini index. This indicates that alterations in brain structure, scores related to cognitive performance, and the clinical characteristics represented by these principal components play a significant role in the advancement of cognitive deterioration. The fifth principal component followed closely (26.3%), PC4 (23.8%), PC1 (21.7%), and Gender (5.2%).

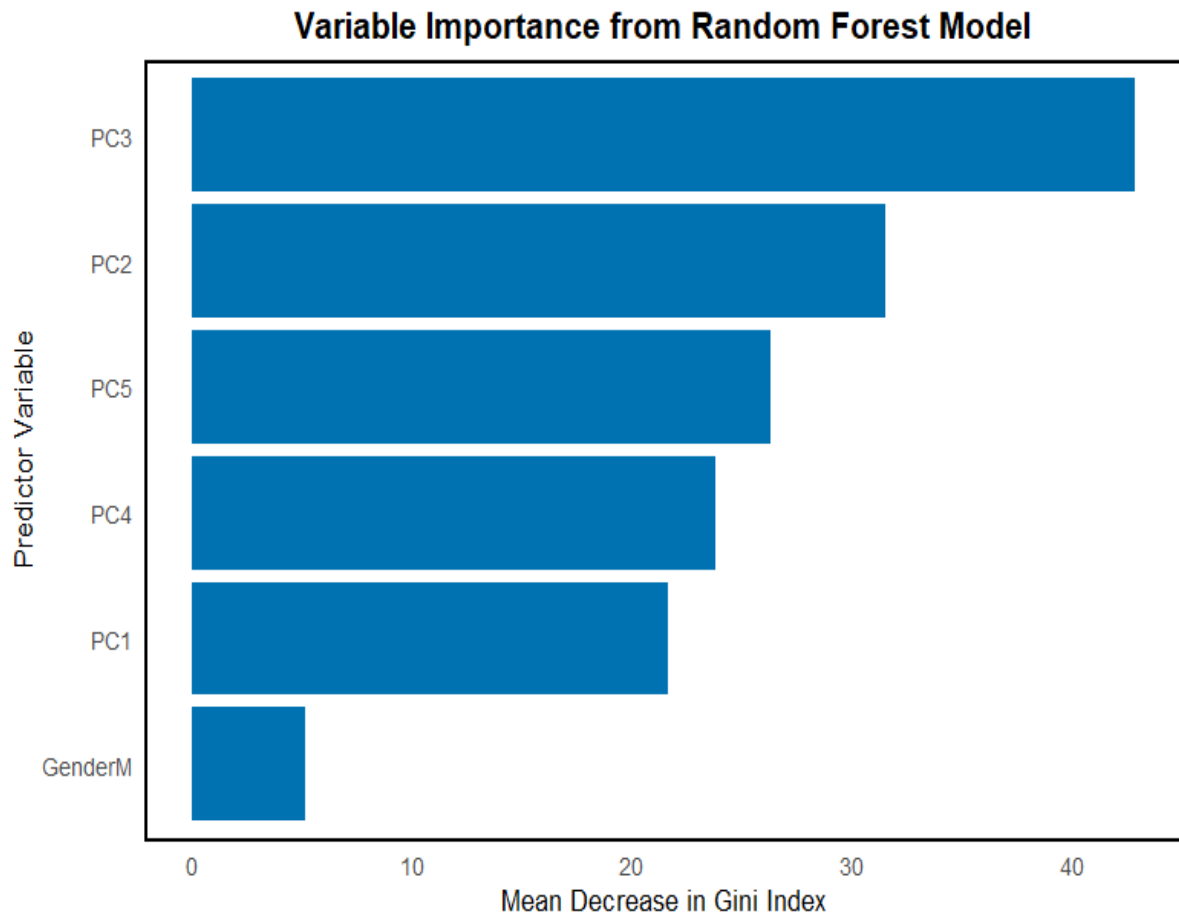


Fig. 4.16: Variable importance analysis

4.9 Analysis in Connection to Prior Research

The results of this research closely align with the contributing factors for predicting dementia and/or Alzheimer's disease discussed in the Literature Review section. Alzheimer's disease arises from a complex interplay of clinical demographics, brain-related traits, and factors impacting cognitive function, rather than a single element. The predictive models developed in this study reflected these interactions and confirmed many associations identified in previous studies.

- **Cognitive Function scores**

Previous research (Grundman et al., 2002; Mouton et al., 1998) has consistently demonstrated the importance of cognitive evaluation tools like the MMSE and CDR as effective indicators of dementia severity in individuals with mild cognitive impairment (MCI), functioning as a predictive measure for the potential progression to Alzheimer's disease (AD). This aligns closely with the analysis conducted in this study. The selected predictive model for Alzheimer's disease (AD) revealed the third principal component as the most important predictor, which is heavily loaded with the MMSE variable.

- **Brain-Related traits**

The findings of this study support previous research on the importance of brain characteristics, including both normalized total brain volume and calculated intracranial volume. In agreement with the work of Apostolova et al. (2012) and Jack Jr. et al. (2000), reductions in brain size indicate the progression of Alzheimer's disease or variations in existing anatomy, as evidenced by the significance of the third and fifth principal components, which are highly correlated with nWBV and eTIV, respectively. Nonetheless, the inverse relationship observed between the severity of dementia and nWBV closely corresponds with previous research on the Atrophy-Cognition model, which suggests that major declines in critical brain regions, such as hippocampal atrophy, result in comparable cognitive symptoms.

- **Clinical Demographics**

The research findings reaffirm previous literature on the role of clinical demographics in predicting and categorizing the progression of dementia. The impact of socioeconomic status and education level, as evidenced through the selected model's variable importance analysis highlighting the significance of the second principal

component, and predominantly influenced by these two factors, further supports the cognitive reserve hypothesis introduced by Stern (2007), suggesting that individuals with higher educational attainment tend to experience a delayed onset or a more gradual progression of dementia symptoms.

Furthermore, this study supports the research methods commonly applied in the prediction modeling of Alzheimer's disease. While conventional statistical techniques such as Multivariate Analysis of Variance, correlation, and regression analysis have played a crucial role in examining differences in brain structures between individuals with dementia and those without, as well as in associating cognitive scores with clinical and neuroimaging markers of cognitive decline (Cole et al., 1993; Liu-Seifert et al., 2016), they face constraints due to their assumptions of linearity, independence, and balanced class distributions. These constraints limit their effectiveness in capturing the intricate, nonlinear relationships present in actual crash data, as evidenced in our findings from the longitudinal OASIS dataset utilized in this research.

The machine learning models utilized in this study, especially Random Forest and Gradient Boosting, demonstrated superior predictive accuracy compared to conventional methods. In line with the results from Genuer et al. (2010) and Fabris et al. (2018), Random Forest exhibited the highest level of predictive accuracy, highlighting the algorithm's effectiveness in addressing imbalanced datasets in our study, thus lowering the risk of misclassifying subjects with severe dementia when compared to other machine learning classifiers used here. These findings are consistent with the work of Song et al. (2021), who showed that Random Forest predicts AD more reliably using fewer features.

It is essential to acknowledge that this study, in summary, adds to the existing literature by showing that machine learning techniques, specifically Random Forest and Gradient Boosting, can significantly enhance the predictive accuracy of models assessing the progression of

Alzheimer's disease, especially in scenarios where conventional statistical methods struggle to address non-linear and complex relationships among demographic, clinical, and structural brain factors. Furthermore, the findings highlight the intricate, multifactorial aspects of Alzheimer's disease progression, which collectively influence the variations observed in dementia severity.

4.10 Discussion and Interpretation of Findings

The predictive models created in this research to forecast and classify the severity of dementia provided valuable insights into the factors contributing to the disease. The ensemble methods, especially Gradient Boosting and Random Forest, outperformed individual learning techniques such as Neural Networks, Support Vector Machines, and Naive Bayes, highlighting their ability to build a collection of classifiers that vote on predictions to enhance their forecasting capabilities. These results support existing research and address the complex non-linear interactions, interdependencies, and significant class imbalances that are typical in clinical and biomedical datasets. The relatively poor performance of these specific learning algorithms highlights the constraints noted in the existing literature. Their challenges in managing non-linearities and imbalanced datasets, such as those faced by neural networks and naive Bayes, along with the meticulous kernel adjustments needed for SVMs, often result in interpretative difficulties.

The analysis of the research indicated once more that various factors, such as age, normalized whole-brain volume, estimated total intracranial volume, level of education, individual socioeconomic status, and Mini-Mental State Examination scores, play a crucial role in precisely modeling the progression of Alzheimer's disease. The analysis of variable importance underscored these factors due to their strong correlation with the principal components retained for modeling disease severity. Gender had a partial influence during the predictive modeling

process; hence, sex-based differences might not significantly impact progression when considering clinical and imaging characteristics. This finding somewhat contradicts existing literature, as women are generally more prone to developing dementia, primarily attributed to their higher life expectancy in old age (Andrew & Tierney, 2018).

The failure of the K-means clustering technique to yield clear, distinct, and non-overlapping subgroups supports the notion proposed in prior studies, suggesting that Alzheimer's disease is perceived as a progressive continuum ranging from asymptomatic preclinical stages to complete dementia, rather than a series of separate clinical phases. This observation, however, contradicts the views held by Butler Hospital (2023) and the Alzheimer's Association (2025), which have developed staging models that categorize the disease into distinct, well-defined clinical stages critical for advancing effective disease-modifying therapies (DMTs).

To summarize, the use of preprocessing methods, such as the Synthetic Minority Over-sampling Technique and Principal Component Analysis, effectively addressed the significant class imbalance and streamlined the dataset's dimensions, thereby enhancing its interpretability. The integration of these preprocessing strategies with ensemble learning models presents a coherent approach for predicting the advancement of dementia. The results of this research highlight the significant contributions that machine learning techniques, particularly Random Forest and Gradient Boosting, make to modeling the progression of Alzheimer's disease within the specified study framework. Nonetheless, they also emphasize the necessity for hybrid models that combine the interpretability of traditional statistical methods with the predictive capabilities of machine learning.

CHAPTER FIVE

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATION

5.1 Summary of Findings

The primary objective of this research was to investigate the gradual changes in brain structure and cognitive impairment associated with Alzheimer's disease. Specifically, the study utilized a combination of statistical techniques and machine learning algorithms to predict and categorize the severity of the disease, while also assessing the effectiveness of these approaches.

The analysis indicated that the dominant factors contributing to Alzheimer's disease severity included Age, Socioeconomic Status, MMSE, normalized Whole Brain Volume (nWBV), and estimated Total Intracranial Volume (eTIV) across all five predictive models, as shown by the Variable Importance Analysis via the predictive significance of the retained principal components' correlation with these variables. Consistent with the existing literature, as clearly stated in Chapter Two of this study, this pattern reveals the magnitude of cognitive function scores, global brain features, and clinical demographics in analyzing dementia severity. In view of this, however, the significance of gender disparities in classifying and predicting the severity of the disease must not be overlooked, as evidenced by its minimal importance in the analysis, as well as previous studies. One plausible explanation for the prevalence of male dementia cases compared to females could be the lower number of documented dementia cases in relation to non-demented ones, alongside a significant gender imbalance.

Additionally, the results from the comparative modeling indicated notable variations in predictive performance between the ensemble methods and the standalone learning techniques across the five predictive models. The standalone learning techniques—Support Vector

Machines, Neural Networks, and Naive Bayes—exhibited relatively lower performance, with accuracy ranging from 73.6% to 77.7% and AUC scores that were generally beneath those of the ensemble methods, although all models faced challenges with misclassification due to significant class imbalance, even after applying the SMOTE resampling technique. On the other hand, the ensemble methods—Random Forest and Gradient Boosting—achieved accuracy exceeding 90% and AUC scores that were typically higher. Interestingly, the Neural Networks delivered the best discriminative capability with an AUC of 91.9% and significantly fewer misclassifications, especially for cases of moderate and severe dementia. Random Forest closely followed with an AUC of 88.8%, also showcasing robust predictive performance and effective management of class imbalance.

The key discoveries indicate that integrating neuroimaging with clinical information enhances the capability to predict and categorize the severity of dementia, as well as to forecast Alzheimer's disease progression, providing a dependable structure for early diagnosis and ongoing disease assessment.

5.2 Conclusion

The findings of this research underscore the significance of cognitive evaluations, total brain volumes assessed through magnetic resonance imaging (MRI), and clinical variables such as socioeconomic status, age, and gender in the assessment of Alzheimer's disease progression and the classification of dementia severity, supporting previous studies that emphasize these factors as essential determinants of dementia results. It is anticipated that the knowledge acquired from this study will be relevant to clinical practices and neuroscience approaches aimed at preventing the disease and enhancing health, particularly given the absence of a globally recognized therapeutic cure for Alzheimer's disease. Understanding the progression of Alzheimer's disease, along with classifying and anticipating the severity of dementia cases,

is crucial for assessing and identifying the risk factors of the condition. This foundation will support additional research focused on identifying elements that may lower exposure to these disease-related factors. Furthermore, it will facilitate the prioritization of high-risk patients for specific interventions.

Among the models evaluated, standalone learning techniques demonstrated the lowest effectiveness, exhibiting a limited ability to manage nonlinear interactions and class imbalance. Ensemble methods yielded moderate results. The models' inclination to overfit, thereby decreasing their applicability to new data, was somewhat alleviated by Principal Component Analysis, which also helped lessen multicollinearity through the dataset's dimensionality reduction.

Overall, the results highlight the capacity of predictive modeling as a valuable decision-support instrument for prioritizing high-risk patients in relation to particular interventions. With additional enhancements, better class equilibrium, and the incorporation of larger, more extensive datasets, machine learning techniques like Random Forest and Gradient Boosting could be utilized by policymakers and highway authorities to actively detect high-risk situations and enact focused measures to lessen the progression of Alzheimer's disease, thereby reducing the severity of dementia.

5.3 Recommendations

Based on the findings and conclusions of this study;

- It is essential to prioritize the clinical implementation of ensemble-based predictive models. Healthcare facilities and neuroscientists should focus on utilizing ensemble classification techniques, such as Random Forest and Gradient Boosting, rather than relying solely on individual learning methods when developing and deploying effective

clinical decision-support systems. This approach will ultimately improve both primary and secondary prevention of the disease by accurately identifying patients at elevated risk of neurodegeneration, thereby enhancing early intervention and monitoring efforts.

- Enhancing data quality and ensuring class balance are essential. Institutions that gather data on dementia and Alzheimer's disease should work on correcting class imbalances to better predict and classify the severity of dementia cases, while minimizing the need for SMOTE resampling, particularly for the severely affected cases that were underrepresented in this research. Including more variables in the dataset, such as genetic and environmental variations in prevalence and biomarkers, the Global Deterioration Scale (GDS), along with behavioral and lifestyle factors, would further enhance model accuracy.
- Well-designed surveillance systems need to be established on both global and national scales to track disease prevalence and the risk factors linked to it. This effort will ultimately help gauge the severity of the disease and its progression, enabling the evaluation of various strategies aimed at reducing the disease burden.

5.4 Limitations of the Study

Although this study met its goals, it faced specific limitations. The dataset utilized was either simulated or public, and may not accurately represent the complexities of dementia in real life. Consequently, the results could vary if the models were applied to different datasets, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. Furthermore, the issue of class imbalance posed a difficulty, as there were fewer occurrences of severe dementia cases, complicating the models' ability to predict high-severity outcomes effectively. Lastly, although this study examined five modeling techniques, a variety of other advanced algorithms could be investigated in future research to enhance predictive accuracy.

5.5 Suggestion for Future Research

Drawing from the current findings, subsequent research should aim to apply the models created in this study to Alzheimer's disease data sourced from both primary and secondary databases. This will provide more robust validation of the models' predictive capabilities and practical significance. Additionally, it is essential to use larger and more varied datasets that feature participants from different ethnicities, education levels, and socioeconomic statuses to improve the models' generalizability and ensure that the predictive frameworks are relevant across various populations and clinical settings. Furthermore, research is needed that extends beyond mere prediction to assess the impact of actual policy interventions in mitigating cognitive decline.

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APPENDICES

R Codes

```
### Import Required Libraries
```

```
library(readxl)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(ggcorrplot)
```

```
library(tidyverse)
```

```
library(MVN)
```

```
library(biotools)
```

```
library(effectsize)
```

```
library(psych)
```

```
library(factoextra)
```

```
library(plotly)
```

```
library(cluster)
```

```
library(caret)
```

```
library(gbm)
```

```
library(dplyr)
```

```
library(pROC)
```

```
library(MLmetrics)
```

```
library(smotefamily)
```

```
library(randomForest)
```

```
library(e1071)
```

```
library(nnet)
```

```
library(naivebayes)
```

```
library(car)
```

```
library(gridExtra)
```



```
### Data Importation
```

```
OASIS_data <- read_excel("C:\\Users\\HP\\Desktop\\project\\oasis_longitudinal.xlsx")
```

```
quant_vars <- c("Visit", "MR Delay", "Age", "EDUC", "SES", "MMSE", "CDR", "eTIV", "nWBV",  
"ASF")
```

```
## Handling Missing Values
```

```
OASIS_data <- OASIS_data %>%
```

```
  filter(!is.na(MMSE), !is.na(SES))
```

```
##### Exploratory Data Analysis #####
```

```
## 1. Descriptive statistics
```

```
summary_stats <- data.frame(  
  Variable = quant_vars,
```

```
  Mean = round(sapply(OASIS_data[, quant_vars], mean, na.rm = TRUE), 2),
```

```
  Median = round(sapply(OASIS_data[, quant_vars], median, na.rm = TRUE), 2),
```

```
  SD = round(sapply(OASIS_data[, quant_vars], sd, na.rm = TRUE), 2),
```

```
  Q1 = round(sapply(OASIS_data[, quant_vars], quantile, probs = 0.25, na.rm = TRUE), 2),
```

```
  Q3 = round(sapply(OASIS_data[, quant_vars], quantile, probs = 0.75, na.rm = TRUE), 2),
```

```
  Min = round(sapply(OASIS_data[, quant_vars], min, na.rm = TRUE), 2),
```

```
  Max = round(sapply(OASIS_data[, quant_vars], max, na.rm = TRUE), 2)
```

```
)
```

```
print(summary_stats)
```

```
## Dementia Status
```

```
ggplot(OASIS_data, aes(x = Group, fill = Group)) +
```

```
  geom_bar(color = "black", width = 0.7) +
```

```
  labs(title = "Distribution of Dementia Status",
```

```
        x = "Dementia Status", y = "Count") +
```

```
  theme_bw() +
```

```

theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid = element_blank(),
  panel.border = element_rect(color = "black", fill = NA, size = 1),
  axis.line = element_line(color = "black"),
  legend.position = "none",
  plot.margin = ggplot2::margin(0, 5, -10, 5, unit = "pt")
) +
scale_y_continuous(expand = expansion(mult = c(0, 0.08)))

## Dementia status Vs Gender
ggplot(OASIS_data, aes(x = `M/F`, fill = Group)) +
  geom_bar(color = "black", width = 0.7, position = "dodge") +
  labs(title = "Dementia Status by Gender", x = "Gender", y = "Count") +
  scale_fill_manual(values = c("#99ff99", "#66b3ff", "#ff9999")) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    panel.grid = element_blank(),
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_line(color = "black"),
    legend.position = c(0.89, 0.85),
    legend.key.size = unit(0.2, "cm"),
    legend.background = element_rect(color = "black", fill = "white"),
    legend.text = element_text(size = 7),
    plot.margin = ggplot2::margin(0, 5, -15, 5, unit = "pt")
  ) +
scale_y_continuous(expand = expansion(mult = c(0, 0.02)))

```

```
## Dementia Status by Age Group
```

```
OASIS_data$AgeGroup <- cut(OASIS_data$Age,
  breaks = c(60, 70, 80, 90, 100),
  labels = c("60-70", "71-80", "81-90", "91-100"),
  right = FALSE,
  include.lowest = TRUE)

ggplot(OASIS_data, aes(x = AgeGroup, fill = Group)) +
  geom_bar(color = "black", width = 0.7, position = "dodge") +
  labs(title = "Dementia Status by Age Group", x = "Age Group (years)", y = "Count") +
  scale_fill_manual(values = c("#99ff99", "#66b3ff", "#ff9999")) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    panel.grid = element_blank(),
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_line(color = "black"),
    legend.position = c(0.88, 0.85),
    legend.key.size = unit(0.2, "cm"),
    legend.background = element_rect(color = "black", fill = "white"),
    legend.text = element_text(size = 8),
    plot.margin = ggplot2::margin(0, 5, -15, 5, unit = "pt")
  ) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.03)))
```

```
## Dementia Status Vs MMSE Scores
```

```
ggplot(OASIS_data, aes(x = Group, y = MMSE, fill = Group)) +
  geom_boxplot(color = "black", alpha = 0.8) +
```

```

labs(title = "Distribution of MMSE Scores by Dementia Status",
      x = "Dementia Status", y = "MMSE Score") +
scale_fill_manual(values = c("#99ff99", "#66b3ff", "#ff9999")) +
theme_bw() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "white"),
  panel.border = element_rect(color = "black", fill = NA),
  legend.position = "none",
  plot.margin = ggplot2::margin(0, 5, -15, 5, unit = "pt")
)

```

CDR Scores by Dementia Status

```

ggplot(OASIS_data, aes(x = Group, y = CDR, fill = Group)) +
geom_boxplot(color = "black", alpha = 0.8) +
labs(title = "Distribution of CDR Scores by Dementia Status",
      x = "Dementia Status", y = "CDR Score") +
scale_fill_manual(values = c("#99ff99", "#66b3ff", "#ff9999")) +
theme_bw() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid = element_blank(),
  panel.border = element_rect(color = "black", fill = NA),
  legend.position = "none",
  plot.margin = ggplot2::margin(0, 5, -15, 5, unit = "pt")
)

```

```

## Dementia Status by Socioeconomic Status

ggplot(na.omit(OASIS_data), aes(x = as.factor(SES), fill = Group)) +
  geom_bar(color = "black", width = 0.7, position = "dodge") +
  labs(title = "Dementia Status by Socioeconomic Status (SES)",
       x = "Socioeconomic Status", y = "Count") +
  scale_fill_manual(values = c("#99ff99", "#66b3ff", "#ff9999")) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    panel.grid = element_blank(),
    panel.background = element_rect(fill = "white"),
    panel.border = element_rect(color = "black", fill = NA),
    axis.line = element_line(color = "black"),
    legend.position = c(0.87, 0.82),
    legend.key.size = unit(0.2, "cm"),
    legend.background = element_rect(color = "black", fill = "white"),
    legend.text = element_text(size = 8),
    plot.margin = ggplot2::margin(0, 5, -15, 5, unit = "pt")
  ) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.03)))

```

```

## Dementia Status Vs Education Level

```

```

OASIS_data$EducGroup <- cut(OASIS_data$EDUC,
  breaks = c(0, 10, 15, 20, 30),
  labels = c("6–10", "11–15", "16–20", "21–25"),
  include.lowest = TRUE, right = FALSE)

ggplot(OASIS_data, aes(x = EducGroup, fill = Group)) +

```

```

geom_bar(color = "black", width = 0.7, position = "dodge") +
labs(title = "Dementia Status by Education Level",
      x = "Education (Years)", y = "Count") +
scale_fill_manual(values = c("#99ff99", "#66b3ff", "#ff9999")) +
theme_bw() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "white"),
  panel.border = element_rect(color = "black", fill = NA),
  axis.line = element_line(color = "black"),
  legend.position = c(0.87, 0.82),
  legend.key.size = unit(0.2, "cm"),
  legend.background = element_rect(color = "black", fill = "white"),
  legend.text = element_text(size = 8),
  plot.margin = ggplot2::margin(0, 5, -15, 5, unit = "pt")
) +
scale_y_continuous(expand = expansion(mult = c(0, 0.03)))

```

nWBV Vs Dementia Status

```

ggplot(OASIS_data, aes(x = Group, y = nWBV, fill = Group)) +
geom_boxplot(color = "black") +
theme_bw() +
theme(
  panel.grid = element_blank(),
  plot.title = element_text(face = "bold", hjust = 0.5),
  legend.position = "none"
) +

```

```

labs(
  title = "Normalized Whole Brain Volume by Dementia Status",
  x = "Dementia Status",
  y = "nWBV"
) +
scale_fill_manual(values = c("skyblue", "salmon", "palegreen3"))

## Correlation matrix
num_vars <- OASIS_data %>%
  select(Age, EDUC, SES, MMSE, CDR, eTIV, nWBV, ASF)

cor_matrix <- cor(num_vars, use = "complete.obs", method = "pearson")

## Correlation Heatmap
ggcorrplot(cor_matrix,
  method = "square",
  type = "full",
  lab = TRUE,
  lab_size = 3,
  hc.order = TRUE,
  colors = c("red", "white", "blue"),
  title = "Correlation Heatmap of MRI, Demographic, and Cognitive Variables")

###MANOVA Assumption Testing
manova_data <- OASIS_data %>%
  select(Group, MMSE, eTIV, nWBV)

mvn_result <- mvn(

```

```

data = OASIS_data[, c("MMSE", "nWBV", "eTIV")],
desc = TRUE
)

by(OASIS_data$MMSE, OASIS_data$Group, shapiro.test)
by(OASIS_data$nWBV, OASIS_data$Group, shapiro.test)
by(OASIS_data$eTIV, OASIS_data$Group, shapiro.test)

OASIS_subset <- OASIS_data[, c("MMSE", "eTIV", "nWBV")]

OASIS_subset <- na.omit(OASIS_subset)
mahal <- mahalanobis(OASIS_subset,
                     colMeans(OASIS_subset),
                     cov(OASIS_subset))

OASIS_subset$Mahalanobis <- mahal
cutoff <- qchisq(0.999, df = ncol(OASIS_subset))
OASIS_subset$Outlier <- mahal > cutoff
table(OASIS_subset$Outlier)

boxM(manova_data[, 2:4], manova_data$Group)

##RM MANOVA
RM_MANOVA <- manova(cbind(MMSE, nWBV, eTIV) ~ Group * Visit, data = OASIS_data)
summary(RM_MANOVA, test = "Pillai")

summary.aov(RM_MANOVA)

```



```

aov_MMSE <- aov(MMSE ~ Group * Visit, data = OASIS_data)
summary(aov_MMSE)

aov_nWBV <- aov(nWBV ~ Group * Visit, data = OASIS_data)
summary(aov_nWBV)

aov_eTIV <- aov(eTIV ~ Group * Visit, data = OASIS_data)
summary(aov_eTIV)

eta_squared(aov_MMSE, partial = TRUE)
eta_squared(aov_nWBV, partial = TRUE)
eta_squared(aov_eTIV, partial = TRUE)

## Principal Component Analysis

OASIS_PCA_data <- OASIS_data[, c("Visit", "MR Delay", "Age", "EDUC", "SES", "MMSE", "eTIV",
"nWBV")]

OASIS_PCA_data <- na.omit(OASIS_PCA_data)

OASIS_PCA_scaled <- scale(OASIS_PCA_data)

PCA_result <- prcomp(OASIS_PCA_scaled, center = TRUE, scale. = TRUE)
summary(PCA_result)

fviz_eig(PCA_result, addlabels = TRUE, ylim = c(0, 60))

PCA_loadings <- PCA_result$rotation[, 1:5]
round(PCA_loadings, 3)

## PCA Visualization

vars_for_pca <- c("Visit", "MR Delay", "Age", "EDUC", "SES", "MMSE", "eTIV", "nWBV")
idx <- complete.cases(OASIS_data[, vars_for_pca])
pca_df <- as.data.frame(PCA_result$x[, 1:5])

```

```

pca_df$Group <- as.factor(OASIS_data$Group[idx])
pca_df <- subset(pca_df, Group != "Converted")
plot_ly(
  data = pca_df,
  x = ~PC1, y = ~PC2, z = ~PC3,
  color = ~Group,
  colors = c("#1f77b4", "#ff7f0e", "#2ca02c"),
  type = "scatter3d",
  mode = "markers",
  marker = list(size = 5, opacity = 0.8)
) %>%
  layout(
    title = "3D PCA Visualization by Group",
    scene = list(
      xaxis = list(title = "PC1"),
      yaxis = list(title = "PC2"),
      zaxis = list(title = "PC3")
    ),
    legend = list(
      x = 0.8, y = 0.85,
      bgcolor = "rgba(255,255,255,0.7)",
      bordercolor = "black",
      borderwidth = 0.5
    )
  )

```

K-Means Clustering

```
pca_scores <- as.data.frame(PCA_result$x[, 1:5])
```

```

## Elbow Method for Optimal k

set.seed(123)

wss <- sapply(1:10, function(k){
  kmeans(pca_scores, centers = k, nstart = 25)$tot.withinss
})

plot(1:10, wss, type = "b", pch = 19, frame = FALSE,
     xlab = "Number of Clusters (k)",
     ylab = "Total Within-Cluster Sum of Squares",
     main = "Elbow Method for Optimal k")

## Silhouette Analysis

set.seed(123)

fviz_nbclust(pca_scores, kmeans, method = "silhouette", k.max = 10)

## Cluster Visuaization

kmeans_result <- kmeans(PCA_result$x[, 1:5], centers = 5, nstart = 25)

pca_df <- as.data.frame(PCA_result$x[, 1:5])
colnames(pca_df)[1:5] <- paste0("PC", 1:5)
pca_df$Cluster <- as.factor(kmeans_result$cluster)

centroids <- as.data.frame(kmeans_result$centers)
colnames(centroids)[1:5] <- paste0("PC", 1:5)
centroids$Cluster <- factor(1:5)

ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster)) +

```

```

geom_point(size = 3, alpha = 0.8) +
geom_point(data = centroids, aes(x = PC1, y = PC2),
           color = "black", fill = "white", size = 6, shape = 21, stroke = 1.2) +
geom_text(data = centroids, aes(x = PC1, y = PC2, label = Cluster),
          color = "black", size = 4, vjust = 0.5, fontface = "bold") +
labs(
  title = "K-Means Clustering (k = 5)",
  x = "Principal Component 1",
  y = "Principal Component 2",
  color = "Cluster"
) +
scale_color_manual(values = c("#1f77b4", "#2ca02c", "#ff7f0e", "#FFD700", "#9467bd")) +
theme_minimal(base_size = 14) +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid = element_blank(),
  panel.border = element_rect(color = "black", fill = NA, linewidth = 1.1),
  plot.margin = ggplot2::margin(15, 20, 10, 20, unit = "pt"),
  legend.position = "right"
) +
coord_cartesian(clip = "off") +
scale_x_continuous(expand = c(0.02, 0)) +
scale_y_continuous(expand = c(0.02, 0))

```

```
## Silhouette Coefficient
```

```

sil <- silhouette(kmeans_result$cluster, dist(PCA_result$x[, 1:5]))
mean_silhouette <- mean(sil[, 3])
cat("Average Silhouette Score:", mean_silhouette, "\n")

```

```
### Model to forecast and classify dementia progression
```

```
##=====Gradient Boosting=====##
```

```
## Data preparation
```

```
OASIS_clean <- na.omit(OASIS_data)
```

```
pca_data <- as.data.frame(PCA_result$x[, 1:5])
```

```
colnames(pca_data) <- paste0("PC", 1:5)
```

```
pca_data$CDR <- as.character(OASIS_clean$CDR)
```

```
pca_data$CDR <- gsub("\\.", "_", pca_data$CDR)
```

```
pca_data$CDR <- paste0("CDR", pca_data$CDR)
```

```
pca_data$CDR <- as.factor(pca_data$CDR)
```

```
pca_data$Gender <- as.factor(OASIS_clean$`M/F`)
```

```
## Data partitioning
```

```
set.seed(111)
```

```
index <- createDataPartition(pca_data$CDR, p = 0.7, list = FALSE)
```

```
train_data <- pca_data[index, ]
```

```
test_data <- pca_data[-index, ]
```

```
train_data$Gender <- ifelse(train_data$Gender == "M", 0, 1)
```

```
train_data$CDR <- as.factor(train_data$CDR)
```

```
## SMOTE resampling
```

```
smote_result <- SMOTE(
```

```
  train_data[, !(names(train_data) %in% "CDR")],
```

```
  train_data$CDR,
```

```
  K = 2,
```

```
  dup_size = 3
```

```

)

train_smote <- smote_result$data

colnames(train_smote)[ncol(train_smote)] <- "CDR"

train_smote$CDR <- as.factor(train_smote$CDR)

train_smote$Gender <- as.numeric(as.character(train_smote$Gender))

train_smote$Gender <- factor(ifelse(train_smote$Gender >= 0.5, "F", "M"))


## Cross-validation control

fitControl <- trainControl(

  method = "repeatedcv",

  number = 5,

  repeats = 3,

  classProbs = TRUE,

  summaryFunction = multiClassSummary,

  savePredictions = "final"

)


## Tuning

gbmGrid <- expand.grid(

  n.trees = c(500, 1000, 1500),

  interaction.depth = c(3, 5, 7),

  shrinkage = c(0.01, 0.05, 0.1),

  n.minobsinnode = c(5, 10)

)


## Model training

set.seed(111)

gbm_tuned <- train(

```

```

CDR ~ PC1 + PC2 + PC3 + PC4 + PC5 + Gender,

data = train_smote,

method = "gbm",

trControl = fitControl,

tuneGrid = gbmGrid,

metric = "Accuracy",

verbose = FALSE

)

## Cross-validation results

gbm_preds <- gbm_tuned$pred %>%

  mutate(pred = as.factor(pred), obs = as.factor(obs))

f1_scores <- gbm_preds %>%

  group_by(Resample) %>%

  summarise(F1 = tryCatch({

    MLmetrics::F1_Score(y_pred = pred, y_true = obs,

      positive = as.character(unique(obs)[1]))

  }, error = function(e) NA)) %>%

  pull(F1)

gbm_cv_results <- data.frame(

  Model = "Gradient Boosting",

  CV_Accuracy_Mean = round(mean(gbm_tuned$resample$Accuracy, na.rm = TRUE), 3),

  CV_Accuracy_SD = round(sd(gbm_tuned$resample$Accuracy, na.rm = TRUE), 3),

  CV_F1_Mean = round(mean(f1_scores, na.rm = TRUE), 3),

  CV_F1_SD = round(sd(f1_scores, na.rm = TRUE), 3)

)

```

```

## Independent test results

pred_classes <- predict(gbm_tuned, newdata = test_data)
pred_classes <- factor(pred_classes, levels = levels(train_smote$CDR))
test_data$CDR <- factor(test_data$CDR, levels = levels(train_smote$CDR))
cm_test <- confusionMatrix(pred_classes, test_data$CDR)

precision_macro_test <- mean(cm_test$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro_test  <- mean(cm_test$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro_test      <- mean(cm_test$byClass[, "F1"], na.rm = TRUE)
test_accuracy      <- as.numeric(cm_test$overall["Accuracy"])
gbm_cv_results$Test_Accuracy <- round(test_accuracy, 3)

## Full dataset evaluation

full_data <- rbind(train_smote, test_data)
full_pred <- predict(gbm_tuned, newdata = full_data)
full_pred <- factor(full_pred, levels = levels(full_data$CDR))
full_data$CDR <- factor(full_data$CDR, levels = levels(full_data$CDR))
cm_full <- confusionMatrix(full_pred, full_data$CDR)

full_accuracy <- as.numeric(cm_full$overall["Accuracy"])
precision_macro_full <- mean(cm_full$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro_full  <- mean(cm_full$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro_full      <- mean(cm_full$byClass[, "F1"], na.rm = TRUE)

gbm_test_results <- data.frame(
  Model = "Gradient Boosting",
  Accuracy = round(full_accuracy, 3),

```



```

Precision_Macro = round(precision_macro_full, 3),

Recall_Macro = round(recall_macro_full, 3),

F1_Macro = round(f1_macro_full, 3)

)

## Final results

print(gbm_cv_results)

print(gbm_test_results)


##=====Random Forest=====##


## Reuse SMOTE data

set.seed(111)

rf_train_data <- train_smote

rf_train_data$CDR <- droplevels(rf_train_data$CDR)

rf_train_data$Gender <- droplevels(rf_train_data$Gender)


## Cross-validation control

fitControl <- trainControl(

  method = "repeatedcv",

  number = 5,

  repeats = 3,

  classProbs = TRUE,

  summaryFunction = multiClassSummary,

  savePredictions = "final"

)


## Tuning

rfGrid <- expand.grid(

```

```

mtry = c(2, 3, 4, 5)
)

## Model training
set.seed(111)
rf_tuned <- train(
  CDR ~ PC1 + PC2 + PC3 + PC4 + PC5 + Gender,
  data = rf_train_data,
  method = "rf",
  trControl = fitControl,
  tuneGrid = rfGrid,
  metric = "Accuracy",
  ntree = 500,
  importance = TRUE
)

## Cross-validation results
rf_preds <- rf_tuned$pred %>%
  mutate(pred = as.factor(pred), obs = as.factor(obs))

f1_scores <- rf_preds %>%
  group_by(Resample) %>%
  summarise(F1 = tryCatch({
    MLmetrics::F1_Score(y_pred = pred, y_true = obs,
      positive = as.character(unique(obs)[1]))
  }, error = function(e) NA)) %>%
  pull(F1)

rf_cv_results <- data.frame(

```

```

Model = "Random Forest",
CV_Accuracy_Mean = round(mean(rf_tuned$resample$Accuracy, na.rm = TRUE), 3),
CV_Accuracy_SD   = round(sd(rf_tuned$resample$Accuracy, na.rm = TRUE), 3),
CV_F1_Mean       = round(mean(f1_scores, na.rm = TRUE), 3),
CV_F1_SD         = round(sd(f1_scores, na.rm = TRUE), 3)
)

```

```
## Independent test results
```

```

pred_classes <- predict(rf_tuned, newdata = test_data)
pred_classes <- factor(pred_classes, levels = levels(rf_train_data$CDR))
test_data$CDR <- factor(test_data$CDR, levels = levels(rf_train_data$CDR))
cm_test <- confusionMatrix(pred_classes, test_data$CDR)

precision_macro <- mean(cm_test$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro   <- mean(cm_test$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro       <- mean(cm_test$byClass[, "F1"], na.rm = TRUE)
test_accuracy  <- as.numeric(cm_test$overall["Accuracy"])
rf_cv_results$Test_Accuracy <- round(test_accuracy, 3)

```

```
## Full dataset evaluation
```

```

full_data <- rbind(train_smote, test_data)
full_pred <- predict(rf_tuned, newdata = full_data)
full_pred <- factor(full_pred, levels = levels(full_data$CDR))
full_data$CDR <- factor(full_data$CDR, levels = levels(full_data$CDR))
cm_full <- confusionMatrix(full_pred, full_data$CDR)

full_accuracy <- as.numeric(cm_full$overall["Accuracy"])
precision_macro_full <- mean(cm_full$byClass[, "Pos Pred Value"], na.rm = TRUE)

```

```
recall_macro_full <- mean(cm_full$byClass["Sensitivity"], na.rm = TRUE)
f1_macro_full <- mean(cm_full$byClass["F1"], na.rm = TRUE)
```

```
rf_test_results <- data.frame(
  Model = "Random Forest",
  Accuracy = round(full_accuracy, 3),
  Precision_Macro = round(precision_macro_full, 3),
  Recall_Macro = round(recall_macro_full, 3),
  F1_Macro = round(f1_macro_full, 3)
)
```

```
## Final results
```

```
print(rf_cv_results)
print(rf_test_results)
```

```
##=====Support Vector Machine=====##
```

```
## Reuse SMOTE data
```

```
set.seed(111)
```

```
svm_train_data <- train_smote
```

```
svm_train_data$CDR <- droplevels(svm_train_data$CDR)
```

```
svm_train_data$Gender <- droplevels(svm_train_data$Gender)
```

```
## Cross-validation control
```

```
fitControl <- trainControl(
  method = "repeatedcv",
  number = 5,
  repeats = 3,
  classProbs = TRUE,
```

```

summaryFunction = multiClassSummary,
savePredictions = "final"
)

## Tuning
svmGrid <- expand.grid(
  sigma = c(0.001, 0.01, 0.05),
  C = c(0.5, 1, 2, 5)
)

## Model training
set.seed(111)
svm_tuned <- train(
  CDR ~ PC1 + PC2 + PC3 + PC4 + PC5 + Gender,
  data = svm_train_data,
  method = "svmRadial",
  trControl = fitControl,
  tuneGrid = svmGrid,
  metric = "Accuracy"
)

## Cross validation results
svm_preds <- svm_tuned$pred %>%
  mutate(pred = as.factor(pred), obs = as.factor(obs))

f1_scores <- svm_preds %>%
  group_by(Resample) %>%
  summarise(F1 = tryCatch({

```

```

MLmetrics::F1_Score(y_pred = pred, y_true = obs,
                    positive = as.character(unique(obs)[1]))
}, error = function(e) NA)) %>%
pull(F1)

svm_cv_results <- data.frame(
  Model = "Support Vector Machine",
  CV_Accuracy_Mean = round(mean(svm_tuned$resample$Accuracy, na.rm = TRUE), 3),
  CV_Accuracy_SD   = round(sd(svm_tuned$resample$Accuracy, na.rm = TRUE), 3),
  CV_F1_Mean       = round(mean(f1_scores, na.rm = TRUE), 3),
  CV_F1_SD         = round(sd(f1_scores, na.rm = TRUE), 3)
)

## Independent test results
pred_classes <- predict(svm_tuned, newdata = test_data)
pred_classes <- factor(pred_classes, levels = levels(svm_train_data$CDR))
test_data$CDR <- factor(test_data$CDR, levels = levels(svm_train_data$CDR))
cm_test <- confusionMatrix(pred_classes, test_data$CDR)

precision_macro <- mean(cm_test$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro    <- mean(cm_test$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro        <- mean(cm_test$byClass[, "F1"], na.rm = TRUE)
test_accuracy   <- as.numeric(cm_test$overall["Accuracy"])

svm_cv_results$Test_Accuracy <- round(test_accuracy, 3)

## Full dataset evaluation
full_data <- rbind(train_smote, test_data)

```

```

full_pred <- predict(svm_tuned, newdata = full_data)
full_pred <- factor(full_pred, levels = levels(full_data$CDR))
full_data$CDR <- factor(full_data$CDR, levels = levels(full_data$CDR))
cm_full <- confusionMatrix(full_pred, full_data$CDR)

full_accuracy <- as.numeric(cm_full$overall["Accuracy"])
precision_macro_full <- mean(cm_full$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro_full <- mean(cm_full$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro_full <- mean(cm_full$byClass[, "F1"], na.rm = TRUE)

svm_test_results <- data.frame(
  Model = "Support Vector Machine",
  Accuracy = round(full_accuracy, 3),
  Precision_Macro = round(precision_macro_full, 3),
  Recall_Macro = round(recall_macro_full, 3),
  F1_Macro = round(f1_macro_full, 3)
)

## Final results
print(svm_cv_results)
print(svm_test_results)

##=====Neural Network=====##

## Reuse SMOTE data
set.seed(111)
nn_train_data <- train_smote

```

```
nn_train_data$CDR <- droplevels(nn_train_data$CDR)
nn_train_data$Gender <- droplevels(nn_train_data$Gender)
```

```
## Cross-validation control
```

```
fitControl <- trainControl(
  method = "repeatedcv",
  number = 5,
  repeats = 3,
  classProbs = TRUE,
  summaryFunction = multiClassSummary,
  savePredictions = "final"
)
```

```
## Tuning grid
```

```
nnGrid <- expand.grid(
  size = c(3, 5, 7, 9),    # hidden layer nodes
  decay = c(0.001, 0.01, 0.1) # regularization strength
)
```

```
## Model training
```

```
set.seed(111)
nn_tuned <- train(
  CDR ~ PC1 + PC2 + PC3 + PC4 + PC5 + Gender,
  data = nn_train_data,
  method = "nnet",
  trControl = fitControl,
  tuneGrid = nnGrid,
```



```

metric = "Accuracy",
maxit = 500,
trace = FALSE
)

## Cross-validation results

nn_preds <- nn_tuned$pred %>%
  mutate(pred = as.factor(pred), obs = as.factor(obs))

f1_scores <- nn_preds %>%
  group_by(Resample) %>%
  summarise(F1 = tryCatch({
    MLmetrics::F1_Score(y_pred = pred, y_true = obs,
      positive = as.character(unique(obs)[1]))
  }, error = function(e) NA)) %>%
  pull(F1)

nn_cv_results <- data.frame(
  Model = "Neural Network",
  CV_Accuracy_Mean = round(mean(nn_tuned$resample$Accuracy, na.rm = TRUE), 3),
  CV_Accuracy_SD = round(sd(nn_tuned$resample$Accuracy, na.rm = TRUE), 3),
  CV_F1_Mean = round(mean(f1_scores, na.rm = TRUE), 3),
  CV_F1_SD = round(sd(f1_scores, na.rm = TRUE), 3)
)

## Independent test results

pred_classes <- predict(nn_tuned, newdata = test_data)
pred_classes <- factor(pred_classes, levels = levels(nn_train_data$CDR))

```

```

test_data$CDR <- factor(test_data$CDR, levels = levels(nn_train_data$CDR))

cm_test <- confusionMatrix(pred_classes, test_data$CDR)

precision_macro <- mean(cm_test$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro    <- mean(cm_test$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro        <- mean(cm_test$byClass[, "F1"], na.rm = TRUE)
test_accuracy   <- as.numeric(cm_test$overall["Accuracy"])

nn_cv_results$Test_Accuracy <- round(test_accuracy, 3)
nn_cv_results <- nn_cv_results[, c("Model", "Test_Accuracy", "CV_Accuracy_Mean",
                                   "CV_Accuracy_SD", "CV_F1_Mean", "CV_F1_SD")]

## Full dataset evaluation

full_data <- rbind(train_smote, test_data)
full_pred <- predict(nn_tuned, newdata = full_data)
full_pred <- factor(full_pred, levels = levels(full_data$CDR))
full_data$CDR <- factor(full_data$CDR, levels = levels(full_data$CDR))
cm_full <- confusionMatrix(full_pred, full_data$CDR)

full_accuracy <- as.numeric(cm_full$overall["Accuracy"])
precision_macro_full <- mean(cm_full$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro_full    <- mean(cm_full$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro_full        <- mean(cm_full$byClass[, "F1"], na.rm = TRUE)

nn_test_results <- data.frame(
  Model = "Neural Network",
  Accuracy = round(full_accuracy, 3),
  Precision_Macro = round(precision_macro_full, 3),

```

```

Recall_Macro = round(recall_macro_full, 3),
F1_Macro = round(f1_macro_full, 3)
)

## Final results
print(nn_cv_results)
print(nn_test_results)

##=====Naïve Bayes=====##

## Reuse SMOTE data
set.seed(111)
nb_train_data <- train_smote
nb_train_data$CDR <- droplevels(nb_train_data$CDR)
nb_train_data$Gender <- droplevels(nb_train_data$Gender)

## Cross-validation control
fitControl <- trainControl(
  method = "repeatedcv",
  number = 5,
  repeats = 3,
  classProbs = TRUE,
  summaryFunction = multiClassSummary,
  savePredictions = "final"
)

## Model training

```

```

set.seed(111)

nb_tuned <- train(
  CDR ~ PC1 + PC2 + PC3 + PC4 + PC5 + Gender,
  data = nb_train_data,
  method = "naive_bayes",
  trControl = fitControl,
  metric = "Accuracy"
)

## Cross-validation results

nb_preds <- nb_tuned$pred %>%
  mutate(pred = as.factor(pred), obs = as.factor(obs))

f1_scores <- nb_preds %>%
  group_by(Resample) %>%
  summarise(F1 = tryCatch({
    MLmetrics::F1_Score(y_pred = pred, y_true = obs,
      positive = as.character(unique(obs)[1]))
  }, error = function(e) NA)) %>%
  pull(F1)

nb_cv_results <- data.frame(
  Model = "Naïve Bayes",
  CV_Accuracy_Mean = round(mean(nb_tuned$resample$Accuracy, na.rm = TRUE), 3),
  CV_Accuracy_SD = round(sd(nb_tuned$resample$Accuracy, na.rm = TRUE), 3),
  CV_F1_Mean = round(mean(f1_scores, na.rm = TRUE), 3),
  CV_F1_SD = round(sd(f1_scores, na.rm = TRUE), 3)
)

```

```

## Independent test results

pred_classes <- predict(nb_tuned, newdata = test_data)
pred_classes <- factor(pred_classes, levels = levels(nb_train_data$CDR))
test_data$CDR <- factor(test_data$CDR, levels = levels(nb_train_data$CDR))
cm_test <- confusionMatrix(pred_classes, test_data$CDR)

precision_macro <- mean(cm_test$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro <- mean(cm_test$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro <- mean(cm_test$byClass[, "F1"], na.rm = TRUE)
test_accuracy <- as.numeric(cm_test$overall["Accuracy"])

nb_cv_results$Test_Accuracy <- round(test_accuracy, 3)
nb_cv_results <- nb_cv_results[, c("Model", "Test_Accuracy", "CV_Accuracy_Mean",
                                   "CV_Accuracy_SD", "CV_F1_Mean", "CV_F1_SD")]

## Full dataset evaluation (Accuracy Itself)

full_data <- rbind(train_smote, test_data)
full_pred <- predict(nb_tuned, newdata = full_data)
full_pred <- factor(full_pred, levels = levels(full_data$CDR))
full_data$CDR <- factor(full_data$CDR, levels = levels(full_data$CDR))
cm_full <- confusionMatrix(full_pred, full_data$CDR)

full_accuracy <- as.numeric(cm_full$overall["Accuracy"])
precision_macro_full <- mean(cm_full$byClass[, "Pos Pred Value"], na.rm = TRUE)
recall_macro_full <- mean(cm_full$byClass[, "Sensitivity"], na.rm = TRUE)
f1_macro_full <- mean(cm_full$byClass[, "F1"], na.rm = TRUE)

```

```

nb_test_results <- data.frame(
  Model = "Naïve Bayes",
  Accuracy = round(full_accuracy, 3),
  Precision_Macro = round(precision_macro_full, 3),
  Recall_Macro = round(recall_macro_full, 3),
  F1_Macro = round(f1_macro_full, 3)
)

## Final results
print(nb_cv_results)
print(nb_test_results)

##### CV RESULTS #####
CV_RESULTS <- rbind(
  gbm_cv_results,
  rf_cv_results,
  svm_cv_results,
  nn_cv_results,
  nb_cv_results
)

CV_RESULTS[, c("Model", "Test_Accuracy", "CV_Accuracy_Mean",
               "CV_Accuracy_SD", "CV_F1_Mean", "CV_F1_SD")]

##### TEST RESULTS #####
TEST_RESULTS <- rbind(
  gbm_test_results,
  rf_test_results,

```

```

svm_test_results,
nn_test_results,
nb_test_results
)

TEST_RESULTS[, c("Model", "Accuracy", "Precision_Macro",
                 "Recall_Macro", "F1_Macro")]

# === CONFUSION MATRICES === #

# 1. Gradient Boosting
pred_gbm <- predict(gbm_tuned, newdata = test_data)
pred_gbm <- factor(pred_gbm, levels = levels(train_smote$CDR))
gbm_cm <- confusionMatrix(pred_gbm, test_data$CDR)

# 2. Random Forest
pred_rf <- predict(rf_tuned, newdata = test_data)
pred_rf <- factor(pred_rf, levels = levels(train_smote$CDR))
rf_cm <- confusionMatrix(pred_rf, test_data$CDR)

# 3. Support Vector Machine
pred_svm <- predict(svm_tuned, newdata = test_data)
pred_svm <- factor(pred_svm, levels = levels(train_smote$CDR))
svm_cm <- confusionMatrix(pred_svm, test_data$CDR)

# 4. Neural Network
pred_nn <- predict(nn_tuned, newdata = test_data)
pred_nn <- factor(pred_nn, levels = levels(train_smote$CDR))
nn_cm <- confusionMatrix(pred_nn, test_data$CDR)

```

```
# 5. Naïve Bayes
```

```
pred_nb <- predict(nb_tuned, newdata = test_data)
pred_nb <- factor(pred_nb, levels = levels(train_smote$CDR))
nb_cm <- confusionMatrix(pred_nb, test_data$CDR)
```

```
confusion_matrices <- list(
  Gradient_Boosting = gbm_cm,
  Random_Forest = rf_cm,
  SVM = svm_cm,
  Neural_Network = nn_cm,
  Naive_Bayes = nb_cm
)
```

```
plot_conf_matrix <- function(cm, model_name) {
  cm_table <- as.data.frame(cm$table)
  colnames(cm_table) <- c("Predicted", "Actual", "Freq")
  ggplot(cm_table, aes(x = Actual, y = Predicted, fill = Freq)) +
    geom_tile(color = "white") +
    geom_text(aes(label = Freq), color = "black", size = 4) +
    scale_fill_gradient(low = "white", high = "steelblue") +
    theme_minimal(base_size = 12) +
    labs(title = model_name, x = "Actual Class", y = "Predicted Class") +
    theme(plot.title = element_text(hjust = 0.5, face = "bold"))
}
```

```
p1 <- plot_conf_matrix(gbm_cm, "Gradient Boosting")
p2 <- plot_conf_matrix(rf_cm, "Random Forest")
```



```

p3 <- plot_conf_matrix(svm_cm, "Support Vector Machine")
p4 <- plot_conf_matrix(nn_cm, "Neural Network")
p5 <- plot_conf_matrix(nb_cm, "Naïve Bayes")

grid.arrange(p1, p2, p3, p4, p5, ncol = 2)

##### === ROC-AUC === #####

roc_gbm <- multiclass.roc(test_data$CDR, as.numeric(predict(gbm_tuned, newdata = test_data, type =
"prob")[,1]))

roc_rf <- multiclass.roc(test_data$CDR, as.numeric(predict(rf_tuned, newdata = test_data, type =
"prob")[,1]))

roc_svm <- multiclass.roc(test_data$CDR, as.numeric(predict(svm_tuned, newdata = test_data, type =
"prob")[,1]))

roc_nn <- multiclass.roc(test_data$CDR, as.numeric(predict(nn_tuned, newdata = test_data, type =
"prob")[,1]))

roc_nb <- multiclass.roc(test_data$CDR, as.numeric(predict(nb_tuned, newdata = test_data, type =
"prob")[,1]))

auc_values <- data.frame(

  Model = c("Gradient Boosting", "Random Forest", "SVM", "Neural Network", "Naïve Bayes"),

  AUC = c(roc_gbm$auc[1], roc_rf$auc[1], roc_svm$auc[1], roc_nn$auc[1], roc_nb$auc[1])

)

print(auc_values)

roc_plot_data <- data.frame(

  FPR = c(1 - roc_gbm$rocs[[1]]$specificities,

    1 - roc_rf$rocs[[1]]$specificities,

    1 - roc_svm$rocs[[1]]$specificities,

```

```

1 - roc_nn$rocs[[1]]$specificities,
1 - roc_nb$rocs[[1]]$specificities),
TPR = c(roc_gbm$rocs[[1]]$sensitivities,
roc_rf$rocs[[1]]$sensitivities,
roc_svm$rocs[[1]]$sensitivities,
roc_nn$rocs[[1]]$sensitivities,
roc_nb$rocs[[1]]$sensitivities),

Model = factor(rep(c("Gradient Boosting", "Random Forest", "SVM", "Neural Network", "Naïve
Bayes"),

times = c(length(roc_gbm$rocs[[1]]$sensitivities),
length(roc_rf$rocs[[1]]$sensitivities),
length(roc_svm$rocs[[1]]$sensitivities),
length(roc_nn$rocs[[1]]$sensitivities),
length(roc_nb$rocs[[1]]$sensitivities))))

)

ggplot(roc_plot_data, aes(x = FPR, y = TPR, color = Model)) +
geom_line(linewidth = 1.1) +
geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "grey50") +
scale_color_manual(values = c("#0072B2", "#009E73", "#D55E00", "#CC79A7", "#E69F00")) +
labs(
title = "ROC–AUC Curves for All Models",
x = "False Positive Rate (1 - Specificity)",
y = "True Positive Rate (Sensitivity)",
color = "Model"
) +
annotate("text", x = 0.6, y = 0.2, hjust = 0, size = 4, color = "black",
label = paste(
"Gradient Boosting AUC =", round(roc_gbm$auc[1], 3), "\n",

```

```

"Random Forest AUC =", round(roc_rf$auc[1], 3), "\n",
"SVM AUC =", round(roc_svm$auc[1], 3), "\n",
"Neural Net AUC =", round(roc_nn$auc[1], 3), "\n",
"Naïve Bayes AUC =", round(roc_nb$auc[1], 3)
)) +
theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.border = element_rect(color = "black", fill = NA, linewidth = 1.2),
  legend.position = "bottom"
)
##### === Variable Importance === #####
importance_rf <- importance(rf_tuned$finalModel)

importance_rf_df <- data.frame(
  Variable = rownames(importance_rf),
  MeanDecreaseAccuracy = importance_rf[, "MeanDecreaseAccuracy"],
  MeanDecreaseGini = importance_rf[, "MeanDecreaseGini"]
)

importance_rf_df <- importance_rf_df[order(-importance_rf_df$MeanDecreaseGini), ]

print(importance_rf_df)

```